A TREATMENT OF THE CONSTRUCTION OF THE ITÔ INTEGRAL IN THE SPIRIT OF KIYOSI ITÔ'S 1944 AND 1951 PAPERS

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Abstract. These notes present a detailed and self-contained treatment of the construction of the Itô integral and of the formula proved by Kiyosi Itô in *On a Formula Concerning Stochastic Differentials* (Nagoya Math. J. **3** (1951), 55–65), viewed against the background of the earlier 1944 note *Stochastic Integral*. We follow the architecture of the original papers as faithfully as possible, but we write everything in modern notation and with substantially expanded proofs. The integral is first defined on adapted step processes, then extended by an L^2-isometric completion, then enlarged to the class denoted S^* in Itô's paper. After the analytic core has been constructed, we prove the stochastic differential formula in increasing generality: first for polynomial functions, then for smooth compactly supported functions, and finally for the full class of C^2 functions under the growth assumptions needed to justify the approximation argument. Along the way we isolate the structural reasons behind the extra second-order term, namely the nontrivial quadratic variation of Brownian motion. The exposition is intentionally narrative and theorem-driven.

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1. Historical and mathematical orientation

The stochastic integral did not arise as a decorative refinement of classical integration. It arose because the classical chain rule fails when one attempts to compose ordinary smooth functions with Brownian motion. The trajectories of Brownian motion are almost surely continuous and almost surely nowhere differentiable; in particular, the pathwise Riemann–Stieltjes integral with respect to Brownian motion cannot be developed by the ordinary deterministic route. Kiyosi Itô's decisive insight was that the correct notion of integral is not obtained by refining deterministic partitions in a pathwise sense, but rather by exploiting the probabilistic orthogonality of Brownian increments.

The 1944 paper *Stochastic Integral* introduced the integral itself. The 1951 paper *On a Formula Concerning Stochastic Differentials* then supplied the full detailed proof of the differential formula that now bears Itô's name and developed it in a slightly more general setting. In the original notation of Itô, the Brownian motion is written $g(t, \omega)$ and the admissible integrands form classes denoted G, $S(t_0, \ldots, t_n)$, S, $\overline{S}$, and S^*. The 1944 paper is astonishingly short; the mathematical content, however, is deep enough that the short note has supported the whole modern theory of stochastic calculus. The present manuscript expands every step.

Our goals are the following.

(1) To reconstruct the original L^2-theory of the stochastic integral in the spirit of Itô's papers.
(2) To explain why the Itô isometry is the basic structural theorem of the subject.
(3) To prove carefully the formula for stochastic differentials, first in its one-dimensional form and then in the slightly more general differential form appearing in Itô's 1951 treatment.
(4) To keep the argument close to the original logic of the paper while filling in all omitted details and recasting the notation in a modern theorem-proof style.

2. Brownian motion and non-anticipating processes

2.1. **The probabilistic setting.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space carrying a standard Brownian motion

$$B = (B_t)_{0 \le t \le 1}.$$

Thus $B_0 = 0$ almost surely, the sample paths are almost surely continuous, the increments are independent, and for $0 \le s \le t \le 1$,

$$B_t - B_s \sim \mathcal{N}(0, t - s).$$

We write

$$\mathcal{F}_t := \sigma(B_r : 0 \le r \le t)$$

for the natural filtration of B. Whenever needed, we complete the filtration in the usual way, but nothing essential in the L^2 construction depends on this completion.

Remark 2.1. *Itô's 1944 paper phrases the admissibility condition in terms of independence from future increments rather than in the modern language of predictability. Concretely,*

the original condition says that a process $f(t,\omega)$ is non-anticipating if, for each fixed time t, the random objects depending on the past up to time t are independent of each future increment $B_\xi - B_t$, $\xi \geq t$. In modern terms this means that the process is adapted in the sense relevant for the left-point stochastic integral.

2.2. **The Hilbert space.** The natural ambient space is

$$L^2([0,1]\times\Omega) := L^2\big([0,1]\times\Omega, \mathcal{B}([0,1])\otimes\mathcal{F},\, \mathrm{d}t\otimes\mathbb{P}\big),$$

with norm

$$\|f\|_{H^2}^2 := \mathbb{E}\int_0^1 |f_t|^2\,\mathrm{d}t.$$

We shall denote this Hilbert norm simply by $\|f\|_{H^2}$. The notation emphasizes that this is the correct energy norm for the stochastic integral.

3. THE 1951 PAPER AND THE EXACT TARGET FORMULA

The 1951 paper is not merely a restatement of the 1944 construction. It has a sharper goal. In the summary line provided by the journal, Itô explicitly says that the previous paper gave the formula only in outline, and that the purpose of the 1951 article is to prove it in detail in a slightly more general form. This generality concerns both the form of the stochastic differential and the class of test functions to which the formula is applied.

For orientation, it is useful to keep the following three levels separate.

(1) The *construction level*: define $\int_0^t f_s\,\mathrm{d}B_s$ for non-anticipating f.

(2) The *quadratic-variation level*: understand why second-order terms survive.

(3) The *differential-formula level*: prove the stochastic chain rule for functions of processes of the form

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}B_t.$$

The first level is essentially the content of the 1944 paper. The third is the main content of the 1951 paper. The second is the conceptual hinge between them.

The exact modern rendering of the formula toward which all arguments point is the following. If

$$X_t = X_0 + \int_0^t a_s\,\mathrm{d}s + \int_0^t b_s\,\mathrm{d}B_s,$$

and if $F\in C^2(\mathbb{R})$ satisfies the natural integrability assumptions, then

$$F(X_t) - F(X_0) = \int_0^t F'(X_s)b_s\,\mathrm{d}B_s + \int_0^t \Big(F'(X_s)a_s + \frac{1}{2}F''(X_s)b_s^2\Big)\,\mathrm{d}s. \tag{1}$$

A time-dependent version will also be proved below: if $u\in C^{1,2}([0,1]\times\mathbb{R})$, then

$$\begin{aligned} u(t,X_t) - u(0,X_0) &= \int_0^t u_x(s,X_s)b_s\,\mathrm{d}B_s \\ &\quad + \int_0^t \Big(u_t(s,X_s) + a_s u_x(s,X_s) + \frac{1}{2}b_s^2 u_{xx}(s,X_s)\Big)\,\mathrm{d}s. \end{aligned} \tag{2}$$

The longer proof below is organized so that (1) appears first in the spatial case, and (2) is then obtained as a genuine extension rather than as a formal corollary.

4. Discrete probabilistic lemmas behind the construction

The short 1944 note compresses into a few lines several facts that are now standard material in martingale theory. Since the present manuscript aims at a heavy, self-contained treatment, we record the discrete ingredients explicitly.

Lemma 4.1 (Orthogonality of future Brownian increments)**.** *Let $0 \le s \le t \le 1$, and let $\xi \in L^2(\Omega, \mathcal{F}_s, \mathbb{P})$. Then*

$$\mathbb{E}\big[\xi(B_t - B_s)\big] = 0, \qquad \mathbb{E}\big[\xi(B_t - B_s)^2\big] = (t-s)\mathbb{E}[\xi].$$

If in addition $\eta \in L^2(\Omega, \mathcal{F}_s, \mathbb{P})$, then

$$\mathbb{E}\big[\xi\eta(B_t - B_s)\big] = 0.$$

Proof. Since $B_t - B_s$ is independent of $\mathcal{F}_s$ and centered,

$$\mathbb{E}\big[\xi(B_t - B_s)\big] = \mathbb{E}\Big[\xi\,\mathbb{E}(B_t - B_s \mid \mathcal{F}_s)\Big] = 0.$$

Likewise,

$$\mathbb{E}\big[\xi(B_t - B_s)^2\big] = \mathbb{E}\Big[\xi\,\mathbb{E}((B_t - B_s)^2 \mid \mathcal{F}_s)\Big] = (t-s)\mathbb{E}[\xi].$$

The final identity is the same argument with $\xi\eta$ in place of ξ. □

Lemma 4.2 (Orthogonality of stochastic step summands)**.** *Let*

$$Y_i := \xi_{i-1}(B_{t_i} - B_{t_{i-1}}), \qquad i = 1, \dots, n,$$

where each $\xi_{i-1} \in L^2(\Omega, \mathcal{F}_{t_{i-1}}, \mathbb{P})$. Then

$$\mathbb{E}[Y_i Y_j] = 0 \qquad \text{for } i \ne j.$$

Proof. Assume $i < j$. Then Y_i is $\mathcal{F}_{t_{j-1}}$-measurable because it depends only on Brownian increments up to time $t_i \le t_{j-1}$. Hence

$$\mathbb{E}[Y_i Y_j] = \mathbb{E}\Big[Y_i\,\mathbb{E}(Y_j \mid \mathcal{F}_{t_{j-1}})\Big].$$

Now

$$\mathbb{E}(Y_j \mid \mathcal{F}_{t_{j-1}}) = \xi_{j-1}\,\mathbb{E}(B_{t_j} - B_{t_{j-1}} \mid \mathcal{F}_{t_{j-1}}) = 0,$$

so $\mathbb{E}[Y_i Y_j] = 0$. □

Lemma 4.3 (A discrete L^2 maximal inequality)**.** *Let $(M_k)_{k=0}^n$ be a square-integrable martingale with $M_0 = 0$. Then for every $\lambda > 0$,*

$$\mathbb{P}\Big(\max_{0 \le k \le n} |M_k| \ge \lambda\Big) \le \frac{1}{\lambda^2}\mathbb{E}[M_n^2].$$

Proof. Set

$$\tau := \min\{k \in \{0, \dots, n\} : |M_k| \ge \lambda\},$$

with the convention $\tau = n$ if the set is empty. Then τ is a bounded stopping time for the discrete filtration and $(M_{k\wedge\tau})$ is again a square-integrable martingale. Therefore

$$\mathbb{E}[M_{n\wedge\tau}^2] = \sum_{k=1}^{n} \mathbb{E}\big[(M_{k\wedge\tau} - M_{(k-1)\wedge\tau})^2\big] \le \mathbb{E}[M_n^2].$$

On the event $\{\max_{0\le k\le n}|M_k|\ge\lambda\}$, one has $|M_{n\wedge\tau}|=|M_\tau|\ge\lambda$. Hence

$$\lambda^2\mathbb{P}\Big(\max_{0\le k\le n}|M_k|\ge\lambda\Big)\le\mathbb{E}[M_{n\wedge\tau}^2]\le\mathbb{E}[M_n^2].$$

□

Remark 4.4. *Itô phrases the maximal estimate by citing a lemma extending Kolmogorov's inequality. The modern proof above is the martingale form of the same phenomenon. It does not change the content; it merely unwraps the argument in contemporary language.*

5. The step classes $S(t_0,\dots,t_n)$ and S

Definition 5.1 (Elementary adapted step process)**.** *Fix a partition*

$$0=t_0<t_1<\cdots<t_n=1.$$

An elementary adapted step process *subordinate to this partition is a process of the form*

$$f_t(\omega)=\sum_{i=1}^n\xi_{i-1}(\omega)\mathbf{1}_{(t_{i-1},t_i]}(t),$$

where each coefficient ξ_{i-1} belongs to $L^2(\Omega,\mathcal{F}_{t_{i-1}},\mathbb{P})$.

This is the modern counterpart of Itô's class $S(t_0,\dots,t_n)$. Let S denote the union over all partitions.

Remark 5.2. *The adaptedness condition is not decorative. If one replaces each ξ_{i-1} by a variable measurable with respect to a strictly larger sigma-algebra involving future values, then the orthogonality of the stochastic increments is lost and the whole L^2 theory collapses. The entire construction rests on the interplay between left-point evaluation of the integrand and independence of future increments.*

5.1. Definition of the integral on S.

Definition 5.3 (The stochastic integral on S)**.** *Let*

$$f_t=\sum_{i=1}^n\xi_{i-1}\mathbf{1}_{(t_{i-1},t_i]}(t)\in S.$$

For each $t\in[0,1]$, define

$$I_t(f):=\int_0^t f_s\,\mathrm{d}B_s:=\sum_{i=1}^n\xi_{i-1}\big(B_{t_i\wedge t}-B_{t_{i-1}\wedge t}\big). \tag{3}$$

If $t_k\le t<t_{k+1}$, then the sum takes the more explicit form

$$I_t(f)=\sum_{i=1}^k\xi_{i-1}(B_{t_i}-B_{t_{i-1}})+\xi_k(B_t-B_{t_k}),$$

which is precisely the form written by Itô.

Lemma 5.4 (Well-definedness with respect to representation)**.** *The quantity $I_t(f)$ is independent of the particular step representation of f.*

Proof. Suppose that f is written on two different partitions. Refine both partitions to a common refinement

$$0 = s_0 < s_1 < \cdots < s_m = 1.$$

On each interval $(s_{j-1}, s_j]$, the process f is constant with some coefficient $\eta_{j-1} \in L^2(\Omega, \mathcal{F}_{s_{j-1}})$. Both formulas then reduce to

$$\sum_{j=1}^{m} \eta_{j-1}(B_{s_j \wedge t} - B_{s_{j-1} \wedge t}).$$

Thus the definition does not depend on the chosen partition. □

6. Basic properties on the step class

The 1944 paper lists a family of structural properties. We now derive them with complete proofs.

Proposition 6.1 (Linearity)**.** *For $f, g \in S$ and $a, b \in \mathbb{R}$,*

$$I_t(af + bg) = aI_t(f) + bI_t(g), \qquad 0 \le t \le 1.$$

Proof. Choose a common refinement of the step representations of f and g. Then both sides are finite linear combinations of the same Brownian increments, and the claim follows immediately. □

Proposition 6.2 (Normalization)**.** *If $f \equiv 1$, then*

$$I_t(f) = B_t.$$

Proof. Take any partition. Then

$$I_t(1) = \sum_{i=1}^{n} \big(B_{t_i \wedge t} - B_{t_{i-1} \wedge t}\big) = B_t - B_0 = B_t.$$

□

Proposition 6.3 (Path continuity on the step class)**.** *For each $f \in S$, the process $t \mapsto I_t(f)$ has almost surely continuous paths.*

Proof. If $t \in [t_k, t_{k+1}]$, then

$$I_t(f) = \sum_{i=1}^{k} \xi_{i-1}(B_{t_i} - B_{t_{i-1}}) + \xi_k(B_t - B_{t_k}),$$

which is continuous in t because Brownian trajectories are continuous. □

Proposition 6.4 (Mean zero and martingale structure)**.** *For each $f \in S$, the process $(I_t(f))_{0 \le t \le 1}$ is a square-integrable martingale with respect to $(\mathcal{F}_t)_{0 \le t \le 1}$. In particular,*

$$\mathbb{E}[I_t(f)] = 0.$$

Proof. Fix $0 \le s \le t \le 1$. Choose a partition containing s and t. Then the increment $I_t(f) - I_s(f)$ is a finite sum of random variables of the form

$$\eta(B_v - B_u), \qquad u \ge s,$$

where η is $\mathcal{F}_u$-measurable. Since $B_v - B_u$ is independent of $\mathcal{F}_u$ and has mean zero,

$$\mathbb{E}[\eta(B_v - B_u) \mid \mathcal{F}_s] = 0.$$

Summing gives

$$\mathbb{E}[I_t(f) - I_s(f) \mid \mathcal{F}_s] = 0,$$

so $\mathbb{E}[I_t(f) \mid \mathcal{F}_s] = I_s(f)$. Square-integrability follows from the isometry proved below. □

7. The Itô isometry

The key theorem is the L^2 orthogonality relation. It is the real source of the theory.

Theorem 7.1 (Itô isometry on S)**.** *For every $f \in S$ and every $t \in [0,1]$,*

$$\mathbb{E}\,|I_t(f)|^2 = \mathbb{E}\int_0^t |f_s|^2 \,\mathrm{d}s. \tag{4}$$

More generally, if $f, h \in S$, then

$$\mathbb{E}\big[I_t(f)I_t(h)\big] = \mathbb{E}\int_0^t f_s h_s \,\mathrm{d}s. \tag{5}$$

Proof. Write

$$f_s = \sum_{i=1}^n \xi_{i-1}\mathbf{1}_{(t_{i-1},t_i]}(s).$$

Set

$$Y_i := \xi_{i-1}\big(B_{t_i\wedge t} - B_{t_{i-1}\wedge t}\big), \qquad I_t(f) = \sum_{i=1}^n Y_i.$$

Then

$$\mathbb{E}\,|I_t(f)|^2 = \sum_{i=1}^n \mathbb{E}[Y_i^2] + 2\sum_{1\le i<j\le n} \mathbb{E}[Y_iY_j].$$

For the diagonal terms, ξ_{i-1} is $\mathcal{F}_{t_{i-1}}$-measurable, while the increment $B_{t_i\wedge t} - B_{t_{i-1}\wedge t}$ is independent of $\mathcal{F}_{t_{i-1}}$ and has variance $t_i \wedge t - t_{i-1} \wedge t$. Therefore

$$\mathbb{E}[Y_i^2] = \mathbb{E}[\xi_{i-1}^2](t_i \wedge t - t_{i-1} \wedge t).$$

For the cross terms with $i < j$ we condition on $\mathcal{F}_{t_{j-1}}$:

$$\mathbb{E}[Y_iY_j] = \mathbb{E}\big[Y_i\mathbb{E}(Y_j \mid \mathcal{F}_{t_{j-1}})\big].$$

Since

$$\mathbb{E}(Y_j \mid \mathcal{F}_{t_{j-1}}) = \xi_{j-1}\mathbb{E}\big(B_{t_j\wedge t} - B_{t_{j-1}\wedge t} \mid \mathcal{F}_{t_{j-1}}\big) = 0,$$

we obtain $\mathbb{E}[Y_iY_j] = 0$. Hence

$$\mathbb{E}\,|I_t(f)|^2 = \sum_{i=1}^n \mathbb{E}[\xi_{i-1}^2](t_i \wedge t - t_{i-1} \wedge t) = \mathbb{E}\int_0^t |f_s|^2 \,\mathrm{d}s.$$

This proves (4). The bilinear identity (5) follows by polarization from the quadratic identity. □

Corollary 7.2 (Increment isometry)**.** *For $f \in S$ and $0 \le s \le t \le 1$,*

$$\mathbb{E}\,|I_t(f) - I_s(f)|^2 = \mathbb{E}\int_s^t |f_u|^2 \,\mathrm{d}u.$$

Proof. Apply the isometry to $f\mathbf{1}_{(s,t]}$. □

8. Maximal inequality and uniform convergence

The short 1944 note also proves a maximal estimate. In modern language one can derive it from Doob's L^2 maximal inequality, but it is useful to keep the original logic in view.

Theorem 8.1 (Maximal inequality on the step class)**.** *For every $f \in S$ and every $b > 0$,*

$$\mathbb{P}\left\{\sup_{0\le u\le 1} |I_u(f)| \ge b\right\} \le \frac{1}{b^2}\mathbb{E}\int_0^1 |f_s|^2 \,\mathrm{d}s.$$

Proof. By Theorem 6.4, $(I_u(f))_{0\le u\le 1}$ is a square-integrable martingale. Therefore Doob's L^2 inequality gives

$$\mathbb{E}\Big[\sup_{0\le u\le 1} |I_u(f)|^2\Big] \le 4\mathbb{E}\,|I_1(f)|^2 .$$

Combining this with Chebyshev's inequality yields

$$\mathbb{P}\left\{\sup_{0\le u\le 1} |I_u(f)| \ge b\right\} \le \frac{1}{b^2}\mathbb{E}\Big[\sup_{0\le u\le 1} |I_u(f)|^2\Big].$$

This already gives the bound with an extra constant 4. To recover the sharper constant 1 appearing in Itô's note, one may proceed discretely on a partition, apply Kolmogorov's maximal inequality to the orthogonal increment sum, and then pass to the continuous supremum using continuity. Since the original paper obtains the constant 1 by exactly this route, we sketch it.

Let the partition points be $0 = t_0 < \cdots < t_n = 1$ and define

$$S_k := \sum_{i=1}^{k} \xi_{i-1}(B_{t_i} - B_{t_{i-1}}).$$

The increments of (S_k) are orthogonal in L^2, hence Kolmogorov's maximal inequality gives

$$\mathbb{P}\left\{\max_{1\le k\le n} |S_k| \ge b\right\} \le \frac{1}{b^2}\mathbb{E}\,|S_n|^2 .$$

Using the isometry,

$$\mathbb{E}\,|S_n|^2 = \mathbb{E}\int_0^1 |f_s|^2 \,\mathrm{d}s.$$

Now choose refining partitions with mesh tending to zero. Since $I_\cdot(f)$ has continuous paths, the supremum over $[0,1]$ is the limit of the maxima over the partition points. Passing to the limit gives the desired estimate. □

Corollary 8.2 (A Cauchy criterion for uniform convergence in probability)**.** *If $(f^{(n)})$ is a sequence in S such that*

$$\mathbb{E}\int_0^1 \left|f_s^{(n)} - f_s^{(m)}\right|^2 \mathrm{d}s \to 0,$$

then the processes $I_\cdot(f^{(n)})$ form a Cauchy sequence for the uniform norm in probability:

$$\sup_{0\le t\le 1} \left|I_t(f^{(n)}) - I_t(f^{(m)})\right| \to 0 \qquad \textit{in probability.}$$

Proof. Apply Theorem 8.1 to $f^{(n)} - f^{(m)}$. □

9. A full proof of the maximal inequality in Itô's setting

We now revisit the maximal inequality in the exact context of step integrands and derive it carefully from the discrete lemma.

Theorem 9.1 (Maximal inequality for step integrands)**.** *Let $f \in S$ and let*

$$I_t(f) = \int_0^t f_s \, \mathrm{d}B_s.$$

Then for every $b > 0$,

$$\mathbb{P}\Big(\sup_{0\le t\le 1} |I_t(f)| \ge b\Big) \le \frac{1}{b^2}\mathbb{E}\int_0^1 f_s^2 \, \mathrm{d}s.$$

Proof. Choose a representation

$$f_t = \sum_{i=1}^n \xi_{i-1}\mathbf{1}_{(t_{i-1},t_i]}(t).$$

Define the discrete process

$$M_k := \sum_{i=1}^k \xi_{i-1}(B_{t_i} - B_{t_{i-1}}), \qquad k = 0, 1, \dots, n.$$

Then (M_k) is a martingale with respect to $(\mathcal{F}_{t_k})$: indeed,

$$\mathbb{E}(M_k \mid \mathcal{F}_{t_{k-1}}) = M_{k-1} + \xi_{k-1}\mathbb{E}(B_{t_k} - B_{t_{k-1}} \mid \mathcal{F}_{t_{k-1}}) = M_{k-1}.$$

Applying Theorem 4.3,

$$\mathbb{P}\Big(\max_{0\le k\le n} |M_k| \ge b\Big) \le \frac{1}{b^2}\mathbb{E}[M_n^2].$$

By the isometry on S,

$$\mathbb{E}[M_n^2] = \mathbb{E}\int_0^1 f_s^2 \, \mathrm{d}s.$$

So the estimate already holds at the partition points.

It remains to pass from the partition points to all times. If $t \in [t_{k-1}, t_k]$, then

$$I_t(f) = M_{k-1} + \xi_{k-1}(B_t - B_{t_{k-1}}).$$

Hence on each subinterval the path is continuous. Let $(s_m)_{m\ge1}$ be a countable dense subset of $[0, 1]$ containing all partition points. By continuity,

$$\sup_{0\le t\le 1} |I_t(f)| = \sup_{m\ge1} |I_{s_m}(f)| \qquad \text{a.s.}$$

For each N, apply the discrete inequality to the refinement of the partition obtained by adjoining $s_1, \dots, s_N$. The integral process is unchanged, but now the finite set $\{s_1, \dots, s_N\}$ is contained among the refined partition points; therefore

$$\mathbb{P}\Big(\max_{1\le m\le N} |I_{s_m}(f)| \ge b\Big) \le \frac{1}{b^2}\mathbb{E}\int_0^1 f_s^2 \, \mathrm{d}s.$$

Letting $N \to \infty$ and using monotone convergence of events yields the result. □

Corollary 9.2 (Uniform Cauchy criterion from H^2 convergence)**.** *If $(f^{(n)})$ is a sequence in S such that*

$$\sum_{n=1}^{\infty} \left\| f^{(n+1)} - f^{(n)} \right\|_{H^2}^2 < \infty,$$

then the processes $I.(f^{(n)})$ form an almost surely uniformly Cauchy sequence on $[0,1]$.

Proof. Fix $\varepsilon_n := 2^{-n}$. By the preceding theorem,

$$\mathbb{P}\Big(\sup_{0 \le t \le 1} |I_t(f^{(n+1)} - f^{(n)})| > \varepsilon_n \Big) \le 4^n \left\| f^{(n+1)} - f^{(n)} \right\|_{H^2}^2 .$$

By taking a subsequence if necessary in the construction of the completion, one arranges the right-hand side to be summable. Borel–Cantelli then implies that only finitely many of these events occur, which is exactly the uniform Cauchy property. □

10. Extension from S to $\overline{S}$

We now carry out the Hilbert-space completion. This is the conceptual center of the entire theory.

Definition 10.1. *Let $\overline{S}$ denote the closure of S in the Hilbert space $H^2 = L^2([0,1] \times \Omega)$.*

Theorem 10.2 (Extension theorem)**.** *There exists a unique linear map*

$$I_t : \overline{S} \to L^2(\Omega), \qquad f \mapsto I_t(f),$$

for each fixed $t \in [0,1]$, such that the following hold.

(i) *If $f \in S$, then $I_t(f)$ agrees with the previously defined step integral.*

(ii) *For every $f \in \overline{S}$,*

$$\mathbb{E}\, |I_t(f)|^2 = \mathbb{E} \int_0^t |f_s|^2 \, \mathrm{d}s.$$

(iii) *If $f^{(n)} \in S$ and $f^{(n)} \to f$ in H^2, then*

$$I_t(f^{(n)}) \to I_t(f) \qquad \text{in } L^2(\Omega)$$

for each t.

Moreover, the collection $t \mapsto I_t(f)$ admits a continuous version.

Proof. For fixed t, consider the map

$$J_t : S \to L^2(\Omega), \qquad J_t(f) := I_t(f).$$

By Theorem 7.1,

$$\|J_t(f)\|_{L^2(\Omega)}^2 = \mathbb{E} \int_0^t |f_s|^2 \, \mathrm{d}s \le \|f\|_{H^2}^2 .$$

Thus J_t is continuous on S with respect to the H^2 norm and therefore extends uniquely to a continuous linear map on the closure $\overline{S}$.

The isometry passes to the limit by continuity. Indeed, if $f^{(n)} \to f$ in H^2 with $f^{(n)} \in S$, then

$$\left\| I_t(f^{(n)}) - I_t(f^{(m)}) \right\|_{L^2(\Omega)}^2 = \mathbb{E} \int_0^t \left| f_s^{(n)} - f_s^{(m)} \right|^2 \, \mathrm{d}s,$$

so $(I_t(f^{(n)}))_n$ is Cauchy in $L^2(\Omega)$ and hence converges to some limit, which is defined to be $I_t(f)$. The same identity then shows

$$\mathbb{E}\,|I_t(f)|^2 = \mathbb{E}\int_0^t |f_s|^2 \,\mathrm{d}s.$$

It remains to construct a continuous version. Choose $f^{(n)} \in S$ such that

$$\left\| f^{(n+1)} - f^{(n)} \right\|_{H^2} \le 2^{-2n}.$$

By the maximal inequality,

$$\mathbb{P}\left\{ \sup_{0\le t\le 1} \left| I_t(f^{(n+1)}) - I_t(f^{(n)}) \right| > 2^{-n} \right\} \le 2^{2n}\mathbb{E}\int_0^1 \left| f_s^{(n+1)} - f_s^{(n)} \right|^2 \mathrm{d}s \le 2^{-2n}.$$

The series $\sum_n 2^{-2n}$ converges, so by the Borel–Cantelli lemma, almost surely there exists $N(\omega)$ such that for all $n \ge N(\omega)$,

$$\sup_{0\le t\le 1} \left| I_t(f^{(n+1)})(\omega) - I_t(f^{(n)})(\omega) \right| \le 2^{-n}.$$

Hence $(I_\cdot(f^{(n)})(\omega))_n$ is uniformly Cauchy on $[0,1]$ for almost every ω, and therefore converges uniformly to a continuous limit process. This limit is a version of the L^2 family $t \mapsto I_t(f)$ because convergence also holds in L^2 for each fixed t. Uniqueness follows from the uniqueness of limits in L^2. □

Remark 10.3. *This argument is one of the quiet conceptual gems of the original theory. The integral is constructed in the Hilbert space L^2, but continuity in time is recovered by a probabilistic compactness argument rather than inserted by hand.*

11. Locality and enlargement to the class S^*

The 1944 paper goes beyond the closure $\overline{S}$ and defines the integral on a larger class S^* by patching approximations that agree with the target process with arbitrarily large probability.

Definition 11.1 (The class S^*)**.** *A non-anticipating measurable process f belongs to S^* if for every $\varepsilon > 0$ there exists $h \in \overline{S}$ such that*

$$\mathbb{P}\big\{ f_t = h_t \text{ for all } t \in [0,1] \big\} > 1 - \varepsilon.$$

Lemma 11.2 (Locality on $\overline{S}$)**.** *Let $f, h \in \overline{S}$. Suppose there exists a measurable set $A \in \mathcal{F}$ such that*

$$f_t(\omega) = h_t(\omega) \qquad \text{for all } t \in [0,1] \text{ and all } \omega \in A.$$

Then for each t,

$$I_t(f) = I_t(h) \qquad \text{almost surely on } A.$$

Proof. Choose step approximations $f^{(n)}, h^{(n)} \in S$ converging to f, h in H^2 and, by replacing them on A if necessary, arranged so that $f^{(n)} = h^{(n)}$ on A for each n. For step processes the locality statement is immediate from the defining finite sum. Passing to the limit in L^2 gives the claim for $\overline{S}$. □

Theorem 11.3 (Extension to S^*)**.** *For each $f \in S^*$ there exists a process, unique up to indistinguishability, denoted by*

$$I_t(f) = \int_0^t f_s \, \mathrm{d}B_s,$$

with the property that whenever $h \in \overline{S}$ satisfies

$$\mathbb{P}\{f_t = h_t \text{ for all } t \in [0,1]\} > 1 - \varepsilon,$$

then on that same event the integral process of f coincides with that of h.

Proof. For each integer $n \geq 1$, choose $h^{(n)} \in \overline{S}$ such that

$$A_n := \{f_t = h_t^{(n)} \text{ for all } t \in [0,1]\}$$

satisfies $\mathbb{P}(A_n) > 1 - 2^{-n}$. Since $\sum_n \mathbb{P}(A_n^c) < \infty$, the Borel–Cantelli lemma shows that

$$A := \limsup_{n\to\infty} A_n$$

has probability one. For $\omega \in A$, the point ω belongs to infinitely many of the sets A_n. If $\omega \in A_m \cap A_n$, then $h^{(m)} = f = h^{(n)}$ as trajectories on that event, hence by Theorem 11.2

$$I_t(h^{(m)})(\omega) = I_t(h^{(n)})(\omega) \qquad \text{for all } t.$$

Therefore one may define on A

$$I_t(f)(\omega) := I_t(h^{(n)})(\omega)$$

for any n such that $\omega \in A_n$; the value is independent of the choice of n. Since $\mathbb{P}(A) = 1$, this defines the integral almost surely. Uniqueness follows from the same locality argument. □

12. A more explicit proof of the extension to S^*

The passage from $\overline{S}$ to S^* is characteristic of Itô's original exposition and deserves a fully explicit construction. The point is that S^* is not defined as an L^2 closure. Instead it is a pathwise patching class: a process belongs to S^* if it agrees with some step-approximable process with arbitrarily high probability on the whole time interval.

Theorem 12.1 (Construction on S^* by patching)**.** *Let $f \in S^*$. Then there exists a process $I_t(f)$, unique up to indistinguishability, such that:*

(1) for every $\varepsilon > 0$ there exists $h \in \overline{S}$ with

$$\mathbb{P}\{\omega : f_t(\omega) = h_t(\omega) \text{ for all } t \in [0,1]\} > 1 - \varepsilon;$$

(2) on the event of coincidence one has

$$I_t(f) = I_t(h) \qquad \text{for all } t \in [0,1].$$

Proof. Choose $h^{(n)} \in \overline{S}$ such that, with

$$A_n := \{\omega : f_t(\omega) = h_t^{(n)}(\omega) \text{ for all } t \in [0,1]\},$$

we have

$$\mathbb{P}(A_n) > 1 - 2^{-n}.$$

Set

$$A := \limsup_{n\to\infty} A_n = \bigcap_{m=1}^{\infty} \bigcup_{n\geq m} A_n.$$

Since

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n^c) < \infty,$$

Borel–Cantelli implies $\mathbb{P}(A) = 1$.

For $\omega \in A$ and $t \in [0,1]$, choose any index n such that $\omega \in A_n$ and define

$$I_t(f)(\omega) := I_t(h^{(n)})(\omega).$$

This is well-defined. Indeed, if $\omega \in A_n \cap A_m$, then $h^{(n)}(\cdot,\omega) = f(\cdot,\omega) = h^{(m)}(\cdot,\omega)$ for all t, and by locality on $\overline{S}$ one has

$$I_t(h^{(n)})(\omega) = I_t(h^{(m)})(\omega) \qquad \text{for all } t.$$

Thus the value does not depend on the choice of n.

To see uniqueness, suppose $J_t(f)$ is another process with the same patching property. On each A_n one has

$$I_t(f) = I_t(h^{(n)}) = J_t(f) \qquad \text{for all } t,$$

so equality holds on $\limsup A_n$, which has full probability. □

Remark 12.2. *This step is mathematically subtle. In modern textbooks one usually defines the integral directly on predictable square-integrable processes and never introduces S^*. But the 1944 paper genuinely does pass through this larger class. The proof above is therefore not a modern embellishment: it is an expanded version of a structural move already present in Itô's note.*

13. Quadratic variation and why the classical chain rule fails

The extra term in Itô's formula is forced by the quadratic variation of Brownian motion.

Proposition 13.1 (Quadratic variation of Brownian motion)**.** *Let (π_n) be any sequence of partitions of $[0,t]$ whose mesh tends to zero. Then*

$$\sum_{[u,v]\in\pi_n} (B_v - B_u)^2 \to t \qquad \text{in } L^2(\Omega),$$

and hence also in probability.

Proof. Write

$$Q_n := \sum_{[u,v]\in\pi_n} (B_v - B_u)^2.$$

Because the increments are independent and centered Gaussian with variance $v-u$,

$$\mathbb{E}[Q_n] = \sum_{[u,v]\in\pi_n} (v-u) = t.$$

Furthermore,

$$\mathrm{Var}(Q_n) = \sum_{[u,v]\in\pi_n} \mathrm{Var}\big((B_v - B_u)^2\big).$$

If $X \sim \mathcal{N}(0, \sigma^2)$, then $\mathrm{Var}(X^2) = 2\sigma^4$. Thus

$$\mathrm{Var}(Q_n) = 2 \sum_{[u,v]\in\pi_n} (v-u)^2 \le 2\,|\pi_n| \sum_{[u,v]\in\pi_n} (v-u) = 2t|\pi_n| \to 0.$$

Therefore $Q_n \to t$ in L^2. □

Remark 13.2. *If one naïvely wrote* $\mathrm{d}B_t^2 = 0$ *in the deterministic differential style, the chain rule would remain the ordinary one. The proposition shows that this is false in the stochastic regime: the correct infinitesimal mnemonic is not* $(\mathrm{d}B_t)^2 = 0$ *but rather* $(\mathrm{d}B_t)^2 = \mathrm{d}t$ *at the level of second-order terms. The Itô formula is the rigorous incarnation of that mnemonic.*

14. The stochastic differential formula: preparatory lemmas

We now turn to the 1951 paper. Its aim is to prove in detail the formula already announced in 1944. The architecture of the argument is clear: first define the stochastic integral; next prove a formula for finite-difference expansions along partitions; then pass to the limit using the continuity of trajectories, the quadratic variation, and suitable approximation arguments.

14.1. The model semimartingale of the original paper.

For the purposes of Itô's original theorem, one considers processes of the form

$$X_t = X_0 + \int_0^t a_s \,\mathrm{d}s + \int_0^t b_s \,\mathrm{d}B_s, \tag{6}$$

where a and b are non-anticipating processes satisfying square-integrability assumptions strong enough to guarantee that both integrals are well-defined.

The differential notation attached to (6) is

$$\mathrm{d}X_t = a_t \,\mathrm{d}t + b_t \,\mathrm{d}B_t.$$

The formula to be proved is then

$$F(X_t) - F(X_0) = \int_0^t F'(X_s) b_s \,\mathrm{d}B_s + \int_0^t \left(F'(X_s) a_s + \frac{1}{2} F''(X_s) b_s^2 \right) \mathrm{d}s, \tag{7}$$

under the relevant regularity and integrability assumptions on F.

14.2. A deterministic Taylor estimate.

Lemma 14.1 (Second-order Taylor remainder)**.** *Let $F \in C^2(\mathbb{R})$. For every compact interval $K \subset \mathbb{R}$ there exists a modulus of continuity $\omega_K(r) \to 0$ as $r \downarrow 0$ such that for all $x, y \in K$,*

$$F(y) - F(x) = F'(x)(y-x) + \frac{1}{2} F''(x)(y-x)^2 + R(x,y),$$

with

$$|R(x,y)| \le \omega_K(|y-x|)\,(y-x)^2.$$

Proof. By Taylor's theorem with integral remainder,

$$F(y) - F(x) = F'(x)(y-x) + \int_x^y (y-u) F''(u) \,\mathrm{d}u.$$

Subtracting $\frac{1}{2}F''(x)(y-x)^2$ gives

$$R(x,y) = \int_x^y (y-u)\big(F''(u) - F''(x)\big)\,\mathrm{d}u.$$

If $x, y \in K$, then

$$|R(x,y)| \le \sup_{u\in K,\ |u-x|\le|y-x|} \big|F''(u) - F''(x)\big| \int_x^y |y-u|\,\mathrm{d}u.$$

The integral equals $\frac{1}{2}(y-x)^2$ in absolute value, so the claim follows from the uniform continuity of F'' on the compact set K. □

14.3. **Continuity of the model process.**

Lemma 14.2. *If $a \in L^1([0,1]\times\Omega)$ is adapted and $b \in \overline{S}$, then the process defined by (6) admits a continuous version.*

Proof. The finite-variation term $\int_0^t a_s\,\mathrm{d}s$ is continuous in t for every ω outside a null set by absolute continuity. The stochastic integral term has a continuous version by Theorem 10.2. Summing the two versions yields a continuous version of X. □

15. ITÔ'S FORMULA FOR ADAPTED STEP COEFFICIENTS

We begin with a version in which the coefficients are step processes. This is the cleanest place to see how the formula emerges from finite sums.

Theorem 15.1 (Itô's formula for step coefficients)**.** *Suppose*

$$X_t = X_0 + \int_0^t a_s\,\mathrm{d}s + \int_0^t b_s\,\mathrm{d}B_s,$$

where $a, b \in S$ and $F \in C^2(\mathbb{R})$ has bounded first and second derivatives. Then, for all $t \in [0,1]$,

$$F(X_t) - F(X_0) = \int_0^t F'(X_s)b_s\,\mathrm{d}B_s + \int_0^t \left(F'(X_s)a_s + \frac{1}{2}F''(X_s)b_s^2\right)\mathrm{d}s \tag{8}$$

almost surely.

Proof. Fix $t \in [0,1]$ and choose a partition $\pi : 0 = t_0 < \cdots < t_n = t$ refining the step partitions of both a and b. Write $\Delta_i X := X_{t_{i+1}} - X_{t_i}$. Since X is continuous,

$$F(X_t) - F(X_0) = \sum_{i=0}^{n-1}\big(F(X_{t_{i+1}}) - F(X_{t_i})\big).$$

Applying Taylor's theorem to each increment,

$$F(X_{t_{i+1}}) - F(X_{t_i}) = F'(X_{t_i})\Delta_i X + \frac{1}{2}F''(X_{t_i})(\Delta_i X)^2 + R_i.$$

Summing gives

$$F(X_t) - F(X_0) = \sum_i F'(X_{t_i})\Delta_i X + \frac{1}{2}\sum_i F''(X_{t_i})(\Delta_i X)^2 + \sum_i R_i. \tag{9}$$

Because a and b are step processes on the refined partition,

$$\Delta_i X = a_{t_i}(t_{i+1} - t_i) + b_{t_i}(B_{t_{i+1}} - B_{t_i}).$$

Hence

$$\sum_i F'(X_{t_i})\Delta_i X = \sum_i F'(X_{t_i})a_{t_i}(t_{i+1}-t_i) + \sum_i F'(X_{t_i})b_{t_i}(B_{t_{i+1}}-B_{t_i}).$$

These are precisely the Riemann and stochastic sums for

$$\int_0^t F'(X_s)a_s\,\mathrm{d}s \qquad \text{and} \qquad \int_0^t F'(X_s)b_s\,\mathrm{d}B_s.$$

Thus the first-order term already matches the right-hand side of (8).

We now analyze $(\Delta_i X)^2$. Expanding,

$$(\Delta_i X)^2 = a_{t_i}^2(t_{i+1}-t_i)^2 + 2a_{t_i}b_{t_i}(t_{i+1}-t_i)(B_{t_{i+1}}-B_{t_i}) + b_{t_i}^2(B_{t_{i+1}}-B_{t_i})^2.$$

Summing after multiplication by $\frac{1}{2}F''(X_{t_i})$, we must show that the first two pieces vanish in the limit while the last piece converges to

$$\frac{1}{2}\int_0^t F''(X_s)b_s^2\,\mathrm{d}s.$$

Because F'' and a are bounded on the relevant trajectories, the sum of the $(t_{i+1}-t_i)^2$ terms is bounded by a constant times $|\pi|\sum_i(t_{i+1}-t_i) = t|\pi|$, hence tends to 0 as the mesh goes to zero.

For the mixed term, compute its second moment. Set

$$M_\pi := \sum_i F''(X_{t_i})a_{t_i}b_{t_i}(t_{i+1}-t_i)(B_{t_{i+1}}-B_{t_i}).$$

The summands are orthogonal because each coefficient is $\mathcal{F}_{t_i}$-measurable. Hence

$$\mathbb{E}[M_\pi^2] = \sum_i \mathbb{E}\left[F''(X_{t_i})^2 a_{t_i}^2 b_{t_i}^2\right](t_{i+1}-t_i)^3.$$

Since the coefficients are bounded, this is at most $C\sum_i(t_{i+1}-t_i)^3 \le C|\pi|\sum_i(t_{i+1}-t_i)^2 \to 0$. Thus $M_\pi \to 0$ in L^2.

For the quadratic Brownian term, write

$$N_\pi := \sum_i F''(X_{t_i})b_{t_i}^2\big((B_{t_{i+1}}-B_{t_i})^2 - (t_{i+1}-t_i)\big).$$

Again the summands are orthogonal in L^2, because for Gaussian increments the centered squares are independent and mean zero. Therefore

$$\mathbb{E}[N_\pi^2] = 2\sum_i \mathbb{E}\left[F''(X_{t_i})^2 b_{t_i}^4\right](t_{i+1}-t_i)^2 \to 0.$$

Consequently,

$$\sum_i F''(X_{t_i})b_{t_i}^2(B_{t_{i+1}}-B_{t_i})^2 - \sum_i F''(X_{t_i})b_{t_i}^2(t_{i+1}-t_i) \to 0$$

in L^2. The second sum converges to

$$\int_0^t F''(X_s)b_s^2\,\mathrm{d}s$$

by ordinary Riemann integration, since the paths are continuous and the integrand is piecewise continuous.

It remains to treat the remainder term. By Theorem 14.1, on every compact range of the trajectory,

$$|R_i| \le \omega(|\Delta_i X|)\,(\Delta_i X)^2,$$

with $\omega(r) \to 0$ as $r \downarrow 0$. Because X is continuous, $\max_i |\Delta_i X| \to 0$ almost surely as $|\pi| \to 0$. Therefore

$$\sum_i |R_i| \le \omega\left(\max_i |\Delta_i X|\right) \sum_i (\Delta_i X)^2.$$

The factor $\omega(\max_i |\Delta_i X|)$ tends to 0 almost surely, while $\sum_i (\Delta_i X)^2$ remains bounded in probability because its leading contribution is the Brownian quadratic variation term and the lower-order terms vanish. Hence $\sum_i R_i \to 0$ in probability, and by passing to a subsequence if necessary, almost surely. Taking limits in (9) proves (8). □

16. A partition-by-partition proof of Itô's formula

The proof of Theorem 15.1 already contains the core of the argument, but it is worth isolating the bookkeeping in a separate section because this is exactly the computational heart of the 1951 paper. We now perform the Taylor expansion with all pieces displayed and estimated one by one.

Fix a partition

$$\pi : 0 = t_0 < t_1 < \cdots < t_n = t.$$

For each i define

$$\alpha_i := a_{t_i}(t_{i+1} - t_i), \qquad \beta_i := b_{t_i}(B_{t_{i+1}} - B_{t_i}), \qquad \Delta_i X = \alpha_i + \beta_i.$$

Then

$$F(X_{t_{i+1}}) - F(X_{t_i}) = F'(X_{t_i})(\alpha_i + \beta_i) + \frac{1}{2}F''(X_{t_i})(\alpha_i + \beta_i)^2 + R_i.$$

Summing over i gives four principal contributions:

(10)
$$\begin{aligned} F(X_t) - F(X_0) = &\sum_i F'(X_{t_i})\beta_i + \sum_i F'(X_{t_i})\alpha_i \\ &+ \frac{1}{2}\sum_i F''(X_{t_i})\beta_i^2 + \sum_i F''(X_{t_i})\alpha_i\beta_i + \frac{1}{2}\sum_i F''(X_{t_i})\alpha_i^2 + \sum_i R_i. \end{aligned}$$

We analyze the six terms in exactly the order in which they matter.

The stochastic first-order term. The sum

$$\sum_i F'(X_{t_i})\beta_i = \sum_i F'(X_{t_i})b_{t_i}(B_{t_{i+1}} - B_{t_i})$$

is the left-point stochastic sum for the integrand $F'(X_s)b_s$. On step coefficients it is literally the defining expression for

$$\int_0^t F'(X_s)b_s \,\mathrm{d}B_s.$$

The drift first-order term. Likewise,

$$\sum_i F'(X_{t_i})\alpha_i = \sum_i F'(X_{t_i})a_{t_i}(t_{i+1} - t_i) \to \int_0^t F'(X_s)a_s \,\mathrm{d}s,$$

by ordinary Riemann convergence along the continuous trajectory of X.

The quadratic Brownian term. This is the source of the correction:

$$\frac{1}{2}\sum_i F''(X_{t_i})\beta_i^2 = \frac{1}{2}\sum_i F''(X_{t_i})b_{t_i}^2(B_{t_{i+1}} - B_{t_i})^2.$$

Write it as

$$\frac{1}{2}\sum_i F''(X_{t_i})b_{t_i}^2(t_{i+1} - t_i) + \frac{1}{2}N_\pi,$$

where

$$N_\pi := \sum_i F''(X_{t_i})b_{t_i}^2\Big((B_{t_{i+1}} - B_{t_i})^2 - (t_{i+1} - t_i)\Big).$$

The first term is the ordinary Riemann sum for

$$\frac{1}{2}\int_0^t F''(X_s)b_s^2\,\mathrm{d}s.$$

The second goes to zero in L^2, because the centered square increments are pairwise orthogonal and satisfy

$$\mathbb{E}\big[((B_{t_{i+1}} - B_{t_i})^2 - (t_{i+1} - t_i))^2\big] = 2(t_{i+1} - t_i)^2.$$

Hence

$$\mathbb{E}[N_\pi^2] \le 2\sup|F''|^2\sup|b|^4\sum_i(t_{i+1} - t_i)^2 \to 0.$$

The mixed term. The term

$$\sum_i F''(X_{t_i})\alpha_i\beta_i = \sum_i F''(X_{t_i})a_{t_i}b_{t_i}(t_{i+1} - t_i)(B_{t_{i+1}} - B_{t_i})$$

goes to zero in L^2. Indeed, the summands are orthogonal and each has variance of order $(t_{i+1} - t_i)^3$, so the full variance is bounded by a constant times $\sum_i(t_{i+1} - t_i)^3$, which tends to zero as the mesh tends to zero.

The pure drift square term. The term

$$\frac{1}{2}\sum_i F''(X_{t_i})\alpha_i^2$$

is bounded by a constant multiple of $\sum_i(t_{i+1} - t_i)^2$ and therefore vanishes.

The Taylor remainder. Finally,

$$\sum_i R_i$$

vanishes because the oscillation of X on each partition interval tends to zero while the sum of squared increments remains tight. This is precisely the content of Theorem 25.1.

Combining the six analyses yields the stochastic differential formula. In this expanded decomposition one sees transparently why exactly one second-order contribution survives: not the drift square, not the drift-noise cross term, but only the square of the martingale increment.

17. The Itô formula for general L^2 coefficients

We now pass from step coefficients to the natural L^2 completion.

Theorem 17.1 (Itô formula for square-integrable coefficients)**.** *Let*

$$X_t = X_0 + \int_0^t a_s \, \mathrm{d}s + \int_0^t b_s \, \mathrm{d}B_s,$$

where a is adapted with

$$\mathbb{E} \int_0^1 |a_s| \, \mathrm{d}s < \infty,$$

and $b \in \overline{S}$. Assume $F \in C^2(\mathbb{R})$ with bounded first and second derivatives. Then, for all $t \in [0,1]$,

$$F(X_t) - F(X_0) = \int_0^t F'(X_s) b_s \, \mathrm{d}B_s + \int_0^t \left(F'(X_s) a_s + \frac{1}{2} F''(X_s) b_s^2 \right) \mathrm{d}s \tag{11}$$

almost surely.

Proof. Choose $b^{(n)} \in S$ such that $b^{(n)} \to b$ in H^2. Define

$$X_t^{(n)} := X_0 + \int_0^t a_s \, \mathrm{d}s + \int_0^t b_s^{(n)} \, \mathrm{d}B_s.$$

By the maximal inequality,

$$\sup_{0 \le t \le 1} \left| X_t^{(n)} - X_t \right| = \sup_{0 \le t \le 1} \left| \int_0^t (b_s^{(n)} - b_s) \, \mathrm{d}B_s \right| \to 0$$

in probability. Passing to a subsequence, we may assume uniform almost sure convergence on $[0,1]$.

Apply Theorem 15.1 to each $X^{(n)}$:

$$F(X_t^{(n)}) - F(X_0) = \int_0^t F'(X_s^{(n)}) b_s^{(n)} \, \mathrm{d}B_s + \int_0^t \left(F'(X_s^{(n)}) a_s + \frac{1}{2} F''(X_s^{(n)}) (b_s^{(n)})^2 \right) \mathrm{d}s.$$

We now pass term by term to the limit.

Because F is continuous and $X^{(n)} \to X$ uniformly almost surely,

$$F(X_t^{(n)}) \to F(X_t) \qquad \text{almost surely uniformly in } t.$$

The drift term converges by dominated convergence, since F' is bounded and $X_s^{(n)} \to X_s$ almost surely for almost every s.

For the quadratic variation term, decompose

$$F''(X_s^{(n)})(b_s^{(n)})^2 - F''(X_s) b_s^2 = \big(F''(X_s^{(n)}) - F''(X_s) \big) (b_s^{(n)})^2 + F''(X_s) \big((b_s^{(n)})^2 - b_s^2 \big).$$

Since F'' is bounded and continuous, the first term tends to zero in $L^1([0,1] \times \Omega)$ by dominated convergence together with boundedness of $\|b^{(n)}\|_{H^2}$. The second term tends to zero in L^1 because

$$\int_0^1 \left| (b_s^{(n)})^2 - b_s^2 \right| \mathrm{d}s \le \left(\int_0^1 \left| b_s^{(n)} - b_s \right|^2 \mathrm{d}s \right)^{1/2} \left(\int_0^1 \left| b_s^{(n)} + b_s \right|^2 \mathrm{d}s \right)^{1/2}$$

and the second factor is bounded in $L^2(\Omega)$.

For the stochastic term, it is enough to show that

$$F'(X^{(n)})b^{(n)} \to F'(X)b \qquad \text{in } H^2.$$

Indeed, then the Itô isometry gives convergence of the stochastic integrals in L^2. We estimate

$$\left\|F'(X^{(n)})b^{(n)} - F'(X)b\right\|_{H^2} \le \left\|(F'(X^{(n)}) - F'(X))b^{(n)}\right\|_{H^2} + \left\|F'(X)(b^{(n)} - b)\right\|_{H^2}.$$

The second term tends to zero because F' is bounded and $b^{(n)} \to b$ in H^2. For the first term, boundedness of $b^{(n)}$ in H^2 and uniform almost sure convergence of $X^{(n)}$ to X imply convergence to zero by dominated convergence after truncation and standard approximation. Hence the stochastic term converges in L^2.

Passing to the limit proves (11). □

18. From bounded derivatives to the classical C^2 statement

The classical version of Itô's formula is not restricted to bounded derivatives. One assumes enough growth control to make the integrals meaningful. We now prove a convenient form.

Theorem 18.1 (Classical one-dimensional Itô formula)**.** *Let*

$$X_t = X_0 + \int_0^t a_s \,\mathrm{d}s + \int_0^t b_s \,\mathrm{d}B_s,$$

where a is adapted and integrable, $b \in \overline{S}$, and X has a continuous version. Let $F \in C^2(\mathbb{R})$. Assume that for each $t \in [0,1]$,

$$\mathbb{E}\int_0^t \left|F'(X_s)a_s\right| \,\mathrm{d}s < \infty, \qquad \mathbb{E}\int_0^t \left|F''(X_s)\right| b_s^2 \,\mathrm{d}s < \infty, \qquad \mathbb{E}\int_0^t \left|F'(X_s)\right|^2 b_s^2 \,\mathrm{d}s < \infty.$$

Then, for all $t \in [0,1]$,

$$F(X_t) - F(X_0) = \int_0^t F'(X_s)b_s \,\mathrm{d}B_s + \int_0^t \left(F'(X_s)a_s + \frac{1}{2}F''(X_s)b_s^2\right) \mathrm{d}s \tag{12}$$

almost surely.

Proof. Choose a smooth cutoff function $\chi_n \in C_c^\infty(\mathbb{R})$ such that $0 \le \chi_n \le 1$, $\chi_n \equiv 1$ on $[-n, n]$, and $\chi_n \equiv 0$ outside $[-n-1, n+1]$. Let

$$F_n(x) := \chi_n(x)F(x).$$

Then $F_n \in C^2(\mathbb{R})$ and F_n', F_n'' are bounded. By Theorem 17.1,

$$F_n(X_t) - F_n(X_0) = \int_0^t F_n'(X_s)b_s \,\mathrm{d}B_s + \int_0^t \left(F_n'(X_s)a_s + \frac{1}{2}F_n''(X_s)b_s^2\right) \mathrm{d}s.$$

Now let

$$\tau_n := \inf\{s \le t : |X_s| \ge n\} \wedge t.$$

On $[0, \tau_n]$, one has $F_n(X_s) = F(X_s)$, $F_n'(X_s) = F'(X_s)$, and $F_n''(X_s) = F''(X_s)$. Therefore the stopped formula reads

$$F(X_{\tau_n}) - F(X_0) = \int_0^{\tau_n} F'(X_s)b_s \,\mathrm{d}B_s + \int_0^{\tau_n} \left(F'(X_s)a_s + \frac{1}{2}F''(X_s)b_s^2\right) \mathrm{d}s.$$

Since X is continuous, $\tau_n \uparrow t$ almost surely. By the assumed integrability, the drift terms converge by dominated convergence, and the stochastic integral converges in L^2 by the Itô isometry because

$$\mathbb{E}\int_0^t \left|F'(X_s)\right|^2 b_s^2 \,\mathrm{d}s < \infty.$$

Passing to the limit yields (12). □

19. The time-dependent Itô formula

The classical 1951 presentation naturally points toward functions depending on both time and state. We now prove the standard time-dependent extension in the same style as above.

Theorem 19.1 (Time-dependent Itô formula)**.** *Let*

$$X_t = X_0 + \int_0^t a_s \,\mathrm{d}s + \int_0^t b_s \,\mathrm{d}B_s,$$

where a and b satisfy the assumptions of Theorem 18.1. Let

$$u \in C^{1,2}([0,1] \times \mathbb{R})$$

be such that the three processes

$$u_t(s, X_s), \qquad u_x(s, X_s)a_s, \qquad u_{xx}(s, X_s)b_s^2$$

are integrable enough for the expressions below to make sense, and such that

$$\mathbb{E}\int_0^t |u_x(s, X_s)|^2 b_s^2 \,\mathrm{d}s < \infty$$

for each t. Then

$$\begin{aligned} u(t, X_t) - u(0, X_0) = &\int_0^t u_x(s, X_s) b_s \,\mathrm{d}B_s \\ &+ \int_0^t \Big(u_t(s, X_s) + u_x(s, X_s)a_s + \frac{1}{2} u_{xx}(s, X_s) b_s^2 \Big) \,\mathrm{d}s. \end{aligned} \tag{13}$$

Proof. Fix a partition $\pi : 0 = t_0 < \cdots < t_n = t$. Write

$$\Delta_i u := u(t_{i+1}, X_{t_{i+1}}) - u(t_i, X_{t_i}).$$

We split

$$\Delta_i u = \big[u(t_{i+1}, X_{t_{i+1}}) - u(t_i, X_{t_{i+1}})\big] + \big[u(t_i, X_{t_{i+1}}) - u(t_i, X_{t_i})\big].$$

For the first bracket, the mean-value theorem in time gives

$$u(t_{i+1}, X_{t_{i+1}}) - u(t_i, X_{t_{i+1}}) = u_t(\theta_i, X_{t_{i+1}})(t_{i+1} - t_i)$$

for some random $\theta_i \in [t_i, t_{i+1}]$. Summing and using continuity yields convergence to

$$\int_0^t u_t(s, X_s) \,\mathrm{d}s.$$

For the second bracket, apply the one-variable Taylor formula in the spatial variable at time t_i:

$$u(t_i, X_{t_{i+1}}) - u(t_i, X_{t_i}) = u_x(t_i, X_{t_i})\Delta_i X + \frac{1}{2}u_{xx}(t_i, X_{t_i})(\Delta_i X)^2 + R_i.$$

Now exactly the same partition analysis as in the proof of the space-only formula applies, with $u_x(t_i, X_{t_i})$ and $u_{xx}(t_i, X_{t_i})$ in place of $F'(X_{t_i})$ and $F''(X_{t_i})$. Therefore,

$$\sum_i u_x(t_i, X_{t_i})\Delta_i X \to \int_0^t u_x(s, X_s)a_s \,\mathrm{d}s + \int_0^t u_x(s, X_s)b_s \,\mathrm{d}B_s,$$

and

$$\frac{1}{2}\sum_i u_{xx}(t_i, X_{t_i})(\Delta_i X)^2 \to \frac{1}{2}\int_0^t u_{xx}(s, X_s)b_s^2 \,\mathrm{d}s.$$

The remainders vanish as before. Summing the time and space contributions gives (13). □

Remark 19.2. *The time derivative enters linearly because time has finite variation. There is no additional second-order term in the time variable: the symbolic rules remain*

$$(\,\mathrm{d}t)^2 = 0, \qquad \mathrm{d}t\ \mathrm{d}B_t = 0, \qquad (\,\mathrm{d}B_t)^2 = \,\mathrm{d}t.$$

Only the noisy part contributes quadratic variation.

20. Differential notation and the original symbolic rule

It is now legitimate to rewrite Theorem 18.1 in differential form. If

$$\mathrm{d}X_t = a_t \,\mathrm{d}t + b_t \,\mathrm{d}B_t,$$

then

$$\mathrm{d}F(X_t) = F'(X_t)\,\mathrm{d}X_t + \frac{1}{2}F''(X_t)b_t^2 \,\mathrm{d}t.$$

If one formally substitutes $\mathrm{d}X_t = a_t \,\mathrm{d}t + b_t \,\mathrm{d}B_t$, this becomes

$$\mathrm{d}F(X_t) = F'(X_t)a_t \,\mathrm{d}t + F'(X_t)b_t \,\mathrm{d}B_t + \frac{1}{2}F''(X_t)b_t^2 \,\mathrm{d}t.$$

The symbolic multiplication table implicit in this expression is

$$(\,\mathrm{d}t)^2 = 0, \qquad \mathrm{d}t\ \mathrm{d}B_t = 0, \qquad (\,\mathrm{d}B_t)^2 = \,\mathrm{d}t.$$

These are not pointwise algebraic identities. They summarize the limiting behavior of first- and second-order terms along partitions. The first two vanish because they are of higher infinitesimal order; the last survives precisely because Brownian motion has quadratic variation equal to time.

21. A direct proof for $F(x) = x^2$ and the origin of the correction term

A valuable pedagogical checkpoint is the case $F(x) = x^2$. Then Theorem 18.1 becomes

$$X_t^2 - X_0^2 = 2\int_0^t X_s \,\mathrm{d}X_s + \int_0^t b_s^2 \,\mathrm{d}s, \tag{14}$$

where $\mathrm{d}X_s = a_s \,\mathrm{d}s + b_s \,\mathrm{d}B_s$ and

$$\int_0^t X_s \,\mathrm{d}X_s := \int_0^t X_s a_s \,\mathrm{d}s + \int_0^t X_s b_s \,\mathrm{d}B_s.$$

Direct derivation. From the identity

$$X_{t_{i+1}}^2 - X_{t_i}^2 = 2X_{t_i}\Delta_i X + (\Delta_i X)^2,$$

we sum over a partition and pass to the limit. The first sum becomes $2\int_0^t X_s \,\mathrm{d}X_s$. The second converges to $\int_0^t b_s^2 \,\mathrm{d}s$ because the finite-variation square terms vanish and the Brownian square terms converge to time. This is the most transparent manifestation of the Itô correction. □

22. The formula exactly in the spirit of the 1951 paper

We can now package the result in a formulation very close to Itô's 1951 theorem.

Theorem 22.1 (Itô's formula in the original spirit)**.** *Let X be a process of the form*

$$X_t = X_0 + A_t + M_t, \qquad A_t = \int_0^t a_s \,\mathrm{d}s, \qquad M_t = \int_0^t b_s \,\mathrm{d}B_s,$$

where A has absolutely continuous paths and M is the stochastic integral of a non-anticipating square-integrable process. Let $F \in C^2(\mathbb{R})$ satisfy the natural integrability conditions ensuring that the terms below are finite. Then

$$F(X_t) - F(X_0) = \int_0^t F'(X_s)\,\mathrm{d}M_s + \int_0^t F'(X_s)\,\mathrm{d}A_s + \frac{1}{2}\int_0^t F''(X_s)\,\mathrm{d}\langle M\rangle_s,$$

where

$$\langle M\rangle_t = \int_0^t b_s^2 \,\mathrm{d}s.$$

Equivalently,

$$F(X_t) - F(X_0) = \int_0^t F'(X_s) b_s \,\mathrm{d}B_s + \int_0^t \left(F'(X_s)a_s + \frac{1}{2}F''(X_s)b_s^2\right)\mathrm{d}s.$$

Proof. This is just Theorem 18.1, rewritten so as to isolate separately the finite-variation part A and the martingale part M. The notation $\langle M\rangle$ emphasizes that the correction term depends only on the quadratic variation of the martingale component. □

23. Product formula and integration by parts

One immediate but fundamental consequence of Itô's formula is the stochastic product rule. Although this is standard today, it is worth proving in the same theorem-proof style because it clarifies the algebra of quadratic covariation already implicit in the 1951 paper.

Theorem 23.1 (Product formula)**.** *Let*

$$X_t = X_0 + \int_0^t a_s \,\mathrm{d}s + \int_0^t b_s \,\mathrm{d}B_s, \qquad Y_t = Y_0 + \int_0^t c_s \,\mathrm{d}s + \int_0^t d_s \,\mathrm{d}B_s,$$

where all coefficients satisfy the natural square-integrability assumptions. Then

$$X_tY_t - X_0Y_0 = \int_0^t X_s \,\mathrm{d}Y_s + \int_0^t Y_s \,\mathrm{d}X_s + \int_0^t b_s d_s \,\mathrm{d}s. \tag{15}$$

Equivalently,

$$\mathrm{d}(X_tY_t) = X_t \,\mathrm{d}Y_t + Y_t \,\mathrm{d}X_t + b_t d_t \,\mathrm{d}t.$$

Proof. Apply the two-variable time-independent formula to

$$\Phi(x, y) = xy.$$

Then

$$\Phi_x(x, y) = y, \qquad \Phi_y(x, y) = x, \qquad \Phi_{xx} = \Phi_{yy} = 0, \qquad \Phi_{xy} = 1.$$

Since

$$\mathrm{d}X_t = a_t \,\mathrm{d}t + b_t \,\mathrm{d}B_t, \qquad \mathrm{d}Y_t = c_t \,\mathrm{d}t + d_t \,\mathrm{d}B_t,$$

the only second-order contribution is

$$\mathrm{d}X_t \,\mathrm{d}Y_t = b_t d_t \,\mathrm{d}t.$$

Integrating the differential identity yields (15). □

Corollary 23.2 (Quadratic covariation with Brownian motion)**.** *For*

$$M_t = \int_0^t b_s \,\mathrm{d}B_s,$$

one has

$$M_t B_t - M_0 B_0 = \int_0^t M_s \,\mathrm{d}B_s + \int_0^t B_s b_s \,\mathrm{d}B_s + \int_0^t b_s \,\mathrm{d}s.$$

In particular,

$$[M, B]_t = \int_0^t b_s \,\mathrm{d}s.$$

Proof. Apply the preceding theorem with $Y_t = B_t$, that is, with $c \equiv 0$ and $d \equiv 1$. □

24. Interpretation and structural consequences

The preceding theorem has several immediate conceptual consequences.

Proposition 24.1 (Generator identity)**.** *If X solves*

$$\mathrm{d}X_t = a(X_t) \,\mathrm{d}t + \sigma(X_t) \,\mathrm{d}B_t,$$

and if $F \in C^2(\mathbb{R})$ satisfies the relevant integrability assumptions, then

$$F(X_t) - F(X_0) - \int_0^t \left(a(X_s)F'(X_s) + \frac{1}{2}\sigma(X_s)^2 F''(X_s) \right) \mathrm{d}s$$

is a martingale.

Proof. Apply Itô's formula and observe that the remaining term is the stochastic integral

$$\int_0^t F'(X_s)\sigma(X_s) \,\mathrm{d}B_s,$$

which is a martingale. □

Remark 24.2. *This proposition already contains the infinitesimal generator of the diffusion:*

$$\mathcal{L}F(x) = a(x)F'(x) + \frac{1}{2}\sigma(x)^2 F''(x).$$

Thus Itô's differential formula is the local bridge between sample-path stochastic analysis and the analytic theory of parabolic partial differential equations.

25. Appendix: technical details on the remainder term

For completeness we record a clean bound frequently used implicitly in the above arguments.

Lemma 25.1 (Control of the Taylor remainders along partitions)**.** *Let X be a continuous process on $[0,t]$, let $F \in C^2(\mathbb{R})$, and let $\pi = (t_i)$ be a partition of $[0,t]$. Write $\Delta_i X = X_{t_{i+1}} - X_{t_i}$ and define remainders by*

$$F(X_{t_{i+1}}) - F(X_{t_i}) = F'(X_{t_i})\Delta_i X + \frac{1}{2}F''(X_{t_i})(\Delta_i X)^2 + R_i.$$

Then on the event

$$\sup_{0\le s\le t} |X_s| \le R,$$

one has

$$\sum_i |R_i| \le \omega_R\left(\max_i |\Delta_i X|\right)\sum_i (\Delta_i X)^2,$$

where $\omega_R(r) \to 0$ as $r \downarrow 0$.

Proof. Apply Theorem 14.1 on the compact interval $[-R,R]$. □

Remark 25.2 (On the original notation of Itô)**.** *When read alongside the 1944 and 1951 papers, the only substantial notational translation needed is the following. Itô writes the Brownian motion as $g(t,\omega)$, whereas in these notes we write $B_t(\omega)$. His class G corresponds to the family of non-anticipating or adapted processes; his step classes $S(t_0,\dots,t_n)$ and their union S are precisely the adapted square-integrable step processes used here; the closure $\overline{S}$ is the same $L^2([0,1]\times\Omega)$-closure as in the modern Hilbert-space construction; and the class S^* is the enlarged patching class obtained by locality and high-probability agreement with processes already lying in S or $\overline{S}$. Likewise, Itô's notation $I(t,\omega;f)$ is nothing but the stochastic integral $\int_0^t f_s\,\mathrm{d}B_s$, while the properties denoted $(L),(N),(C),(I),(B),(J)$ are, respectively, linearity, normalization, continuity, the isometry, the maximal inequality, and locality. Thus the modern notation used throughout the manuscript changes the typography and shortens the formulas, but it does not alter the logical architecture of the original development.*

26. A near-original reconstruction of the 1951 theorem flow

We now reorganize the results already proved so that the order of presentation follows the 1951 paper much more closely. In that paper, §1 gives definitions and the three-stage construction of the stochastic integral; §2 states five preliminary theorems on continuity, linearity, additivity in the limits of integration, a maximal bound, and stability under approximation; and §3 culminates in Theorem 6, the stochastic differential formula itself. The value of reproducing this order is not merely historical. It reveals that the formula is not an isolated miracle but the terminus of a carefully arranged chain of approximation theorems.

26.1. **Property α and the conditions $(C.1)$, $(C.2)$.** Let $u < v$ and let $\beta = (\beta_t)_{u\le t\le v}$ be a one-dimensional Brownian motion on some measurable set $\Omega_1 \subseteq \Omega$ of full or nearly

full probability. Following Itô, we say that a stochastic process $\xi(t,\omega)$ has *property α with regard to β in $u \le t \le v$* if for each $t \in [u,v]$ the two systems of random variables

$$\{\xi(\tau,\omega) : u \le \tau \le t\} \qquad \text{and} \qquad \{\beta(\sigma,\omega) - \beta(t,\omega) : t \le \sigma \le v\}$$

are independent. In modern language this is a strengthened non-anticipation condition: the past of ξ up to time t is probabilistically separated from the future increments of the driving Brownian motion.

The two hypotheses imposed in §1 of the 1951 paper are then the following.

(C.1) $\xi(t,\omega)$ has property α concerning $\beta(t,\omega)$ in $u \le t \le v$,

(C.2) $\displaystyle\int_u^v \xi(t,\omega)^2 \, \mathrm{d}t < \infty$ for almost all $\omega \in \Omega_1$.

The 1951 construction is then arranged in three cases.

(1) If ξ is uniformly stepwise on a partition $u = s_0 < \cdots < s_n = v$, define the stochastic integral by left-point sums.

(2) If ξ admits a sequence of uniformly stepwise approximants depending only on the past and converging in pathwise L^2_t, define the integral as the almost sure uniform limit of the corresponding stepwise integrals.

(3) In general, truncate by

$$\xi^{(N)}(t,\omega) := \max\{-N, \min\{\xi(t,\omega), N\}\},$$

define the integrals of the truncated processes by Case 2, and then let $N \to \infty$.

This is precisely the route indicated on the first two pages of the 1951 paper, and it is equivalent to the 1944 $S \to \overline{S} \to S^*$ construction once one identifies the pathwise approximability implicit in Case 2 with the closure and patching arguments developed earlier in this manuscript.

26.2. **Case 1 rederived in exact 1951 style.** Suppose that

$$\xi(t,\omega) = \xi(s_{i-1},\omega), \qquad s_{i-1} \le t < s_i,$$

on a partition $u = s_0 < \cdots < s_n = v$. For $u \le s \le t \le v$ and $s_k \le s < s_{k+1}$, $s_\ell \le t < s_{\ell+1}$, the integral is defined by

$$\begin{aligned}\int_s^t \xi(\tau,\omega) \, \mathrm{d}\beta(\tau,\omega) := \sum_{i=k+1}^{\ell} \xi(s_{i-1},\omega)\big(\beta(s_i,\omega) - \beta(s_{i-1},\omega)\big) \\ + \xi(s_k,\omega)\big(\beta(s_{k+1} \wedge t,\omega) - \beta(s,\omega)\big) - \xi(s_\ell,\omega)\big(\beta(s,\omega) - \beta(s_\ell,\omega)\big),\end{aligned}$$

with the obvious simplifications when s or t coincide with partition points. In modern notation this is nothing but

$$\int_s^t \xi_\tau \, \mathrm{d}\beta_\tau = \int_u^t \xi_\tau \, \mathrm{d}\beta_\tau - \int_u^s \xi_\tau \, \mathrm{d}\beta_\tau,$$

where each term on the right is the step-process integral already constructed in §§4.1–25.1. The advantage of restating the 1951 definition is that it makes clear why Theorem 3 below is built into the very syntax of the definition.

27. Theorems 1–5 of the 1951 paper, with full proofs

We now restate and prove, in modern notation but original order, the five preliminary theorems of §2 of the 1951 paper. Each of them has already appeared implicitly above; the purpose here is to show that the 1951 paper is logically complete before the formula itself begins.

Theorem 27.1 (1951 Theorem 1: continuity in the limits of integration)**.** *Assume that ξ satisfies* (C.1) *and* (C.2) *with respect to a Brownian motion β on $[u, v]$. Then for almost every $\omega \in \Omega_1$ the map*

$$(s,t) \longmapsto \int_s^t \xi(\tau,\omega)\ \mathrm{d}\beta(\tau,\omega), \qquad u \le s \le t \le v,$$

is continuous on the simplex $\{(s,t) : u \le s \le t \le v\}$.

Proof. The continuity in the upper limit t follows from the existence of a continuous version of $t \mapsto \int_u^t \xi_\tau\ \mathrm{d}\beta_\tau$, established earlier by the maximal inequality and Borel–Cantelli argument. Once such a version is fixed, continuity in (s,t) is immediate from the identity

$$\int_s^t \xi_\tau\ \mathrm{d}\beta_\tau = \int_u^t \xi_\tau\ \mathrm{d}\beta_\tau - \int_u^s \xi_\tau\ \mathrm{d}\beta_\tau.$$

Indeed the right-hand side is the difference of two continuous functions of s and t. □

Theorem 27.2 (1951 Theorem 2: linearity under joint non-anticipation)**.** *Let ξ and η satisfy* (C.1) *and* (C.2) *with respect to the same Brownian motion β on $[u, v]$, and assume that the pair $\{\xi, \eta\}$ has property α with regard to β. Then for constants $a, b \in \mathbb{R}$ the process $\zeta := a\xi + b\eta$ also satisfies* (C.1) *and* (C.2)*, and for almost all ω,*

$$\int_s^t \zeta(\tau,\omega)\ \mathrm{d}\beta(\tau,\omega) = a\int_s^t \xi(\tau,\omega)\ \mathrm{d}\beta(\tau,\omega) + b\int_s^t \eta(\tau,\omega)\ \mathrm{d}\beta(\tau,\omega).$$

Proof. The L^2 condition is immediate:

$$\int_u^v |a\xi_t + b\eta_t|^2\ \mathrm{d}t \le 2a^2\int_u^v |\xi_t|^2\ \mathrm{d}t + 2b^2\int_u^v |\eta_t|^2\ \mathrm{d}t < \infty$$

almost surely. Property α for ζ follows because the sigma-field generated by the past of ζ is contained in that generated by the pasts of ξ and η, which by hypothesis is independent of future increments of β.

For uniformly stepwise processes the formula is immediate from linearity of the finite sums. For the general case, choose stepwise approximants $\xi^{(n)} \to \xi$ and $\eta^{(n)} \to \eta$ in the sense of the 1951 Case 2 construction, and set $\zeta^{(n)} := a\xi^{(n)} + b\eta^{(n)}$. Then $\zeta^{(n)} \to \zeta$ in $L^2([u, v])$ pathwise, and by stability of the stochastic integral under such approximation,

$$\int_s^t \zeta_\tau^{(n)}\ \mathrm{d}\beta_\tau \longrightarrow \int_s^t \zeta_\tau\ \mathrm{d}\beta_\tau$$

uniformly in (s,t) in probability, while for each n,

$$\int_s^t \zeta_\tau^{(n)}\ \mathrm{d}\beta_\tau = a\int_s^t \xi_\tau^{(n)}\ \mathrm{d}\beta_\tau + b\int_s^t \eta_\tau^{(n)}\ \mathrm{d}\beta_\tau.$$

Passing to the limit yields the result. □

Theorem 27.3 (1951 Theorem 3: additivity in the interval)**.** *Assume that ξ satisfies* (C.1) *and* (C.2)*. Then for every $u \le s_1 \le s_2 \le s_3 \le v$ one has, for almost all ω,*

$$\int_{s_1}^{s_2} \xi(\tau,\omega)\ \mathrm{d}\beta(\tau,\omega) + \int_{s_2}^{s_3} \xi(\tau,\omega)\ \mathrm{d}\beta(\tau,\omega) = \int_{s_1}^{s_3} \xi(\tau,\omega)\ \mathrm{d}\beta(\tau,\omega).$$

Proof. For stepwise processes the defining sums telescope. For general integrands, choose approximants and pass to the limit by uniform convergence on compact time intervals. Equivalently, once the continuous primitive

$$M_t := \int_u^t \xi_\tau\ \mathrm{d}\beta_\tau$$

has been constructed, the identity is simply

$$(M_{s_2} - M_{s_1}) + (M_{s_3} - M_{s_2}) = M_{s_3} - M_{s_1}.$$

□

Theorem 27.4 (1951 Theorem 4: maximal bound)**.** *If ξ satisfies* (1.6) *in Itô's Case 2 construction, then for every $\lambda > 0$,*

$$\mathbb{P}\Big(\sup_{u\le s\le t\le v}\Big|\int_s^t \xi_\tau\ \mathrm{d}\beta_\tau\Big| \ge \lambda\Big) \le \frac{4}{\lambda^2}\,\mathbb{E}\int_u^v \xi_t^2\ \mathrm{d}t.$$

In particular,

$$\mathbb{P}\Big(\sup_{u\le t\le v}\Big|\int_u^t \xi_\tau\ \mathrm{d}\beta_\tau\Big| \ge \lambda\Big) \le \frac{1}{\lambda^2}\,\mathbb{E}\int_u^v \xi_t^2\ \mathrm{d}t.$$

Proof. Apply the L^2 maximal inequality to the martingale

$$M_t := \int_u^t \xi_\tau\ \mathrm{d}\beta_\tau.$$

Then

$$\mathbb{P}\Big(\sup_{u\le t\le v}|M_t| \ge \lambda\Big) \le \frac{1}{\lambda^2}\mathbb{E}[M_v^2] = \frac{1}{\lambda^2}\mathbb{E}\int_u^v \xi_t^2\ \mathrm{d}t.$$

Since

$$\sup_{u\le s\le t\le v}|M_t - M_s| \le 2\sup_{u\le r\le v}|M_r|,$$

the two-parameter estimate follows immediately. □

Theorem 27.5 (1951 Theorem 5: stability under approximation)**.** *Let ξ_n, $n \ge 1$, and ξ satisfy* (C.1) *and* (C.2) *with respect to the same Brownian motion β on $[u,v]$, and assume that the family $\{\xi_n : n \in \mathbb{N}\} \cup \{\xi\}$ has property α with regard to β. If for almost all $\omega \in \Omega_1$,*

$$\int_u^v |\xi_n(t,\omega) - \xi(t,\omega)|^2\ \mathrm{d}t \longrightarrow 0,$$

then

$$\sup_{u\le s\le t\le v}\Big|\int_s^t \xi_n(\tau,\omega)\ \mathrm{d}\beta(\tau,\omega) - \int_s^t \xi(\tau,\omega)\ \mathrm{d}\beta(\tau,\omega)\Big| \longrightarrow 0$$

in probability on Ω_1.

Proof. Set

$$M_t^{(n)} := \int_u^t (\xi_n - \xi)_\tau\ \mathrm{d}\beta_\tau.$$

Then

$$\sup_{u\le s\le t\le v}\left|\int_s^t (\xi_n-\xi)_\tau\ \mathrm{d}\beta_\tau\right| \le 2\sup_{u\le r\le v}|M_r^{(n)}|.$$

Hence by Theorem 27.4,

$$\mathbb{P}\left(\sup_{u\le s\le t\le v}\left|\int_s^t (\xi_n-\xi)_\tau\ \mathrm{d}\beta_\tau\right|>\varepsilon\right)\le \frac{4}{\varepsilon^2}\mathbb{E}\int_u^v |\xi_n(t)-\xi(t)|^2\ \mathrm{d}t.$$

To conclude we must justify that the expectation on the right tends to zero. Since the convergence hypothesis is pathwise almost everywhere and the 1951 paper assumes the family is square-integrable on Ω_1, one applies truncation exactly as Itô does in the proof: first localize to the event on which the L^2_t-norms are uniformly bounded by some M, then invoke dominated convergence there, and finally let the exceptional probability tend to zero. This yields

$$\mathbb{E}\int_u^v |\xi_n(t)-\xi(t)|^2\ \mathrm{d}t\to 0,$$

whence the probability bound tends to zero. □

28. The five lemmas leading to Theorem 6

At the start of §3, Itô introduces an r-dimensional Brownian motion

$$\beta=(\beta^1,\dots,\beta^r)$$

and processes ξ^i, a^i, b^{ij} satisfying

$$\mathrm{d}\xi_t^i = a_t^i\ \mathrm{d}t + b_t^{ij}\ \mathrm{d}\beta_t^j,\qquad i=1,\dots,n,$$

with the usual tensor summation convention in the index j. Before proving the main formula, he isolates a string of lemmas which control iterated stochastic integrals and quadratic products. We reconstruct them here in a way that makes their algebra transparent.

Lemma 28.1 (1951 Lemma 1: stepwise approximation from the past)**.** *Let ξ be a stochastic process on $[u,v]$ such that*

$$\int_u^v \xi(t,\omega)^2\ \mathrm{d}t<\infty$$

for almost all $\omega\in\Omega_1$. Then there exists a sequence of uniformly stepwise processes ξ_n such that, for each time t, the value $\xi_n(t,\omega)$ is measurable with respect to the past of ξ up to time t, and

$$\int_u^v |\xi_n(t,\omega)-\xi(t,\omega)|^2\ \mathrm{d}t\to 0$$

for almost all $\omega\in\Omega_1$.

Proof. Take dyadic partitions of $[u,v]$ and define the left-point step approximation

$$\xi_n(t,\omega):=\xi(t_{k-1}^{(n)},\omega),\qquad t_{k-1}^{(n)}\le t<t_k^{(n)}.$$

Because each value is sampled from the past, the required measurability condition holds. Pathwise L^2 convergence follows from the density of step functions in $L^2([u,v])$ applied separately on almost every sample path for which $t\mapsto\xi(t,\omega)$ belongs to $L^2([u,v])$. □

Lemma 28.2 (1951 Lemma 2: product of two Itô integrals driven by the same Brownian motion)**.** *Let ξ and η satisfy property α with regard to a one-dimensional Brownian motion β on $[u, v]$, and assume*

$$\int_u^v \xi(t,\omega)^2 \, \mathrm{d}t < \infty, \qquad \int_u^v \eta(t,\omega)^2 \, \mathrm{d}t < \infty$$

for almost all $\omega \in \Omega_1$. Then for almost all ω and every $t \in [u, v]$,

$$\left(\int_u^t \xi_s \, \mathrm{d}\beta_s\right)\left(\int_u^t \eta_s \, \mathrm{d}\beta_s\right) = \int_u^t \left(\int_u^s \eta_r \, \mathrm{d}\beta_r\right) \xi_s \, \mathrm{d}\beta_s + \int_u^t \left(\int_u^s \xi_r \, \mathrm{d}\beta_r\right) \eta_s \, \mathrm{d}\beta_s + \int_u^t \xi_s \eta_s \, \mathrm{d}s. \tag{16}$$

Proof. For stepwise processes on a common partition $u = t_0 < \cdots < t_n = v$, write

$$X_k := \sum_{i=1}^k \xi_{i-1} \Delta\beta_i, \qquad Y_k := \sum_{i=1}^k \eta_{i-1} \Delta\beta_i, \qquad \Delta\beta_i := \beta_{t_i} - \beta_{t_{i-1}}.$$

Then the discrete product identity gives

$$X_k Y_k - X_{k-1} Y_{k-1} = X_{k-1} \Delta Y_k + Y_{k-1} \Delta X_k + \Delta X_k \Delta Y_k.$$

Since

$$\Delta X_k = \xi_{k-1} \Delta\beta_k, \qquad \Delta Y_k = \eta_{k-1} \Delta\beta_k, \qquad \Delta X_k \Delta Y_k = \xi_{k-1}\eta_{k-1} (\Delta\beta_k)^2,$$

summing over k yields the stepwise version of (16), except that the last term is a sum with $(\Delta\beta_k)^2$ rather than Δt_k. The difference between the two is exactly the quadratic variation error, which tends to zero in probability because

$$\sum_k \xi_{k-1}\eta_{k-1}\big((\Delta\beta_k)^2 - \Delta t_k\big) \to 0$$

in L^2 by the same orthogonality calculation used in the proof of the Itô isometry. Then approximate general integrands by stepwise ones using Lemma 28.1 and Theorem 27.5. □

Lemma 28.3 (1951 Lemma 3: product of stochastic integrals driven by independent Brownian components)**.** *Let ξ and η satisfy property α with respect to the two-dimensional Brownian motion (β, γ) on $[u, v]$, with the two components independent. Assume both integrands are square-integrable almost surely. Then*

$$\left(\int_u^t \xi_s \, \mathrm{d}\beta_s\right)\left(\int_u^t \eta_s \, \mathrm{d}\gamma_s\right) = \int_u^t \left(\int_u^s \eta_r \, \mathrm{d}\gamma_r\right) \xi_s \, \mathrm{d}\beta_s + \int_u^t \left(\int_u^s \xi_r \, \mathrm{d}\beta_r\right) \eta_s \, \mathrm{d}\gamma_s \tag{17}$$

for almost all ω.

Proof. Repeat the discrete argument from Lemma 28.2, but now

$$\Delta X_k \Delta Y_k = \xi_{k-1}\eta_{k-1} \Delta\beta_k \Delta\gamma_k.$$

Because the Brownian components are independent, the sum of these cross terms converges to zero in L^2: the diagonal expectations vanish, and off-diagonal terms vanish by independence of disjoint future increments. Hence there is no quadratic-variation correction term. □

Lemma 28.4 (1951 Lemma 4: product of a finite-variation integral and a stochastic integral)**.** *Let a and b satisfy property α with respect to a one-dimensional Brownian motion β, and assume*

$$\int_u^v |a_t| \, \mathrm{d}t < \infty, \qquad \int_u^v b_t^2 \, \mathrm{d}t < \infty$$

almost surely. Then

$$\left(\int_u^t a_s \, \mathrm{d}s\right)\left(\int_u^t b_s \, \mathrm{d}\beta_s\right) = \int_u^t a_s \left(\int_u^s b_r \, \mathrm{d}\beta_r\right) \mathrm{d}s + \int_u^t b_s \left(\int_u^s a_r \, \mathrm{d}r\right) \mathrm{d}\beta_s. \tag{18}$$

Proof. Let

$$A_k := \sum_{i=1}^k a_{i-1}\Delta t_i, \qquad M_k := \sum_{i=1}^k b_{i-1}\Delta\beta_i.$$

Then

$$A_k M_k - A_{k-1}M_{k-1} = A_{k-1}\Delta M_k + M_{k-1}\Delta A_k + \Delta A_k \Delta M_k.$$

The first two terms converge to the two integrals in (18). For the last one, note that

$$\sum_i a_{i-1}b_{i-1}\Delta t_i \Delta\beta_i$$

converges to zero in L^2 because

$$\mathbb{E}\left[\left(\sum_i a_{i-1}b_{i-1}\Delta t_i\Delta\beta_i\right)^2\right] = \sum_i \mathbb{E}[a_{i-1}^2 b_{i-1}^2](\Delta t_i)^3 \to 0.$$

□

Lemma 28.5 (1951 Lemma 5: quadratic products of coordinate increments)**.** *Let ξ^i, $i = 1, \ldots, n$, satisfy*

$$\mathrm{d}\xi_t^i = a_t^i \, \mathrm{d}t + b_t^{ij} \, \mathrm{d}\beta_t^j$$

on $[u, v]$, with the usual non-anticipation and integrability hypotheses. Then for each pair of indices i, ℓ and for $u \le t \le s \le v$,

$$\begin{aligned}(\xi_s^i - \xi_t^i)(\xi_s^\ell - \xi_t^\ell) = &\int_t^s \Big((\xi_r^i - \xi_t^i)a_r^\ell + (\xi_r^\ell - \xi_t^\ell)a_r^i + b_r^{ik}b_r^{\ell k}\Big) \, \mathrm{d}r \\ &+ \int_t^s \Big((\xi_r^i - \xi_t^i)b_r^{\ell k} + (\xi_r^\ell - \xi_t^\ell)b_r^{ik}\Big) \, \mathrm{d}\beta_r^k.\end{aligned} \tag{19}$$

Proof. Write

$$\xi_s^i - \xi_t^i = \int_t^s a_r^i \, \mathrm{d}r + \int_t^s b_r^{ik} \, \mathrm{d}\beta_r^k, \qquad \xi_s^\ell - \xi_t^\ell = \int_t^s a_r^\ell \, \mathrm{d}r + \int_t^s b_r^{\ell k} \, \mathrm{d}\beta_r^k.$$

Multiply the two expressions and apply Lemmas 28.2–28.4 term by term. The product of the two stochastic integrals generates the quadratic-variation term $\int_t^s b_r^{ik}b_r^{\ell k} \, \mathrm{d}r$ when the Brownian index agrees, while the mixed $\mathrm{d}\beta^k \, \mathrm{d}\beta^m$ terms with $k \neq m$ disappear by Lemma 28.3. Collecting the terms yields (19). □

29. Theorem 6 exactly in the architecture of the 1951 paper

We can now state the main theorem in a form which matches the original page almost line for line.

Theorem 29.1 (1951 Theorem 6, modernized but in original order)**.** *Let $\beta = (\beta^1, \dots, \beta^r)$ be an r-dimensional Brownian motion on $[u, v]$. Let ξ^i, a^i, b^{ij}, $1 \le i \le n$, $1 \le j \le r$, be stochastic processes such that the family*

$$\{\xi^i, a^i, b^{ij} : 1 \le i \le n,\ 1 \le j \le r\}$$

has property α with respect to β, and suppose

$$\mathrm{d}\xi_t^i = a_t^i \ \mathrm{d}t + b_t^{ij} \ \mathrm{d}\beta_t^j, \qquad 1 \le i \le n.$$

Let $G \subset \mathbb{R}^n$ be open and assume that the range of $t \mapsto \xi_t = (\xi_t^1, \dots, \xi_t^n)$ remains in G almost surely on $[u, v]$. Let

$$f = f(t, x^1, \dots, x^n)$$

be continuous on $[u, v] \times G$, with continuous partial derivatives

$$f_0 := \partial_t f, \qquad f_i := \partial_{x^i} f, \qquad f_{ij} := \partial_{x^i}\partial_{x^j} f.$$

Then the process

$$\eta_t := f(t, \xi_t^1, \dots, \xi_t^n)$$

satisfies

$$\mathrm{d}\eta_t = \Big(f_0(t, \xi_t) + f_i(t, \xi_t) a_t^i + \frac{1}{2} f_{ij}(t, \xi_t) b_t^{ik} b_t^{jk}\Big) \mathrm{d}t + f_i(t, \xi_t) b_t^{ij} \ \mathrm{d}\beta_t^j. \tag{20}$$

Proof. Fix a partition $u = t_0 < t_1 < \cdots < t_m = t$ and write $\Delta_p \xi := \xi_{t_{p+1}} - \xi_{t_p}$. Taylor's formula with second-order remainder gives

$$\begin{aligned} &f(t_{p+1}, \xi_{t_{p+1}}) - f(t_p, \xi_{t_p}) \\ &\quad = f_0(t_p, \xi_{t_p}) \Delta_p t + f_i(t_p, \xi_{t_p}) \Delta_p \xi^i + \frac{1}{2} f_{ij}(t_p, \xi_{t_p}) \Delta_p \xi^i \Delta_p \xi^j + R_p, \end{aligned}$$

where the remainders satisfy

$$\sum_p |R_p| \to 0$$

in probability as the mesh tends to zero, by the same compactness-and-modulus argument developed earlier in Theorem 25.1. Summing over p yields

$$\eta_t - \eta_u = \sum_p f_0(t_p, \xi_{t_p}) \Delta_p t + \sum_p f_i(t_p, \xi_{t_p}) \Delta_p \xi^i + \frac{1}{2} \sum_p f_{ij}(t_p, \xi_{t_p}) \Delta_p \xi^i \Delta_p \xi^j + o_{\mathbb{P}}(1).$$

Insert the differential representation

$$\Delta_p \xi^i = \int_{t_p}^{t_{p+1}} a_r^i \ \mathrm{d}r + \int_{t_p}^{t_{p+1}} b_r^{ij} \ \mathrm{d}\beta_r^j.$$

The first-order sum converges to

$$\int_u^t f_i(r, \xi_r) a_r^i \ \mathrm{d}r + \int_u^t f_i(r, \xi_r) b_r^{ij} \ \mathrm{d}\beta_r^j.$$

For the second-order sum, apply Lemma 28.5 on each subinterval. The finite-variation terms are negligible beyond first order, the mixed stochastic terms vanish in probability,

and only the quadratic-variation contribution survives:

$$\sum_p f_{ij}(t_p, \xi_{t_p})\Delta_p\xi^i\Delta_p\xi^j \longrightarrow \int_u^t f_{ij}(r, \xi_r)b_r^{ik}b_r^{jk}\ \mathrm{d}r.$$

Combining the limits gives

$$\eta_t - \eta_u = \int_u^t f_0(r, \xi_r)\ \mathrm{d}r + \int_u^t f_i(r, \xi_r)a_r^i\ \mathrm{d}r + \int_u^t f_i(r, \xi_r)b_r^{ij}\ \mathrm{d}\beta_r^j + \frac{1}{2}\int_u^t f_{ij}(r, \xi_r)b_r^{ik}b_r^{jk}\ \mathrm{d}r,$$

which is precisely (20) in integrated form. □

30. How the 1944 and 1951 papers fit together exactly

The reconstruction above makes the division of labor between the two classical papers extremely clear.

(1) The 1944 paper provides the integral itself: the classes G, $S(t_0, \ldots, t_n)$, S, S^*; the finite-sum definition; and the six structural properties $(L), (N), (C), (I), (B), (J)$.

(2) The 1951 paper assumes that integral technology and repackages it in a more flexible local form on intervals $[u, v]$, with the condition $(C.1)$ encoding non-anticipation and $(C.2)$ encoding square integrability.

(3) Theorem 1 through Theorem 5 in the 1951 paper are not side remarks; they are the exact stability properties required to manipulate products and Taylor expansions under stochastic integration.

(4) Theorem 6 is then the chain rule for the differential system $\mathrm{d}\xi_t^i = a_t^i\ \mathrm{d}t + b_t^{ij}\ \mathrm{d}\beta_t^j$. The remarkable term

$$\frac{1}{2}f_{ij}(t, \xi_t)b_t^{ik}b_t^{jk}$$

is forced by the quadratic variation of the Brownian components and by nothing else.

In this sense the two papers should be read as one continuous mathematical act: first construct the integral by orthogonality, then use that construction to discover the correct differential calculus for noisy trajectories.

31. Bessel processes and the radial viewpoint

The Itô formula becomes especially vivid when one applies it not merely to an arbitrary smooth observable, but to one that records a geometric feature of Brownian motion. The most classical example is the radial distance of a multidimensional Brownian particle from the origin. If

$$W_t = (W_t^1, \ldots, W_t^d)$$

is a d-dimensional Brownian motion and one sets

$$R_t := |W_t|\,,$$

then R_t measures the distance of the particle from the origin. The remarkable point is that the radial motion is no longer a Brownian motion: the curvature of Euclidean spheres creates a deterministic drift term. The diffusion that appears is the Bessel process.

This is one of the cleanest places where the differential formula explains a phenomenon that is invisible from purely pathwise intuition. In dimension one there is no radial curvature, and $|B_t|$ is obtained from Brownian motion by reflection. In higher dimensions the sphere of radius r has mean curvature of order $1/r$, and this is exactly what reappears as the singular drift in the Bessel equation. In this sense the Bessel process is one of the first genuinely geometric diffusions generated by stochastic calculus.

31.1. **From radial Brownian motion to the Bessel equation.** Let W be a standard d-dimensional Brownian motion started from $x \in \mathbb{R}^d$, let

$$R_t = |W_t|\,, \qquad \tau_0 := \inf\{t \geq 0 : W_t = 0\},$$

and consider the smooth function

$$f(y) = |y| = (y_1^2 + \cdots + y_d^2)^{1/2}$$

on $\mathbb{R}^d \setminus \{0\}$. Its derivatives are

$$\partial_i f(y) = \frac{y_i}{|y|}, \qquad \partial_{ij} f(y) = \frac{\delta_{ij}}{|y|} - \frac{y_i y_j}{|y|^3}, \qquad y \neq 0.$$

Hence

$$\Delta f(y) = \sum_{i=1}^d \partial_{ii} f(y) = \frac{d-1}{|y|}, \qquad y \neq 0.$$

Applying the multidimensional Itô formula up to time $t \wedge \tau_0$ gives

$$R_{t\wedge\tau_0} = |x| + \sum_{i=1}^d \int_0^{t\wedge\tau_0} \frac{W_s^i}{R_s}\,\mathrm{d}W_s^i + \frac{d-1}{2}\int_0^{t\wedge\tau_0} \frac{\mathrm{d}s}{R_s}.$$

The martingale part has quadratic variation

$$\mathrm{d}\left\langle \sum_{i=1}^d \int_0^{\cdot\wedge\tau_0} \frac{W_s^i}{R_s}\,\mathrm{d}W_s^i \right\rangle_t = \sum_{i=1}^d \frac{(W_t^i)^2}{R_t^2}\mathbf{1}_{\{t<\tau_0\}}\,\mathrm{d}t = \mathbf{1}_{\{t<\tau_0\}}\,\mathrm{d}t.$$

Therefore, by Lévy's characterization, there exists a one-dimensional Brownian motion β such that

$$R_{t\wedge\tau_0} = |x| + \beta_{t\wedge\tau_0} + \frac{d-1}{2}\int_0^{t\wedge\tau_0} \frac{\mathrm{d}s}{R_s}.$$

Equivalently, for $t < \tau_0$,

$$\mathrm{d}R_t = \frac{d-1}{2R_t}\,\mathrm{d}t + \mathrm{d}\beta_t.$$

This is the Bessel equation of dimension d.

Definition 31.1. *For $\delta > 0$, a* Bessel process of dimension δ, *denoted* BES^δ, *is a nonnegative diffusion solving*

$$\mathrm{d}R_t = \frac{\delta-1}{2R_t}\,\mathrm{d}t + \mathrm{d}\beta_t, \qquad R_0 = r \geq 0,$$

up to the appropriate boundary behavior at the origin.

The parameter δ is called the dimension parameter because when $\delta = d \in \{1, 2, 3, \dots\}$ the process is the radial part of d-dimensional Brownian motion. The equation

$$\mathrm{d}Z_t = \frac{a}{Z_t}\,\mathrm{d}t + \mathrm{d}W_t$$

corresponds to the Bessel dimension

$$\delta = 2a + 1.$$

Thus the threshold $a = 1/2$ is exactly the threshold $\delta = 2$ that separates the regime where the origin is hit from the regime where the origin is not reached.

31.2. **The squared Bessel process and the generator.** One of the cleanest structural facts is obtained by squaring the radial process.

Proposition 31.2 (Squared Bessel equation)**.** *If R is a BES^δ process, then*

$$X_t := R_t^2$$

satisfies

$$\mathrm{d}X_t = 2\sqrt{X_t}\,\mathrm{d}\beta_t + \delta\,\mathrm{d}t, \qquad X_0 = r^2.$$

Conversely, if X is a nonnegative solution of this equation, then $R_t := \sqrt{X_t}$ is a Bessel process of dimension δ away from the origin.

Proof. Apply Itô's formula to the function $x \mapsto x^2$. Since

$$\mathrm{d}R_t = \frac{\delta - 1}{2R_t}\,\mathrm{d}t + \mathrm{d}\beta_t, \qquad (\mathrm{d}R_t)^2 = \mathrm{d}t,$$

we obtain

$$\mathrm{d}(R_t^2) = 2R_t\,\mathrm{d}R_t + \mathrm{d}t = 2R_t\,\mathrm{d}\beta_t + (\delta - 1)\,\mathrm{d}t + \mathrm{d}t.$$

Hence

$$\mathrm{d}X_t = 2\sqrt{X_t}\,\mathrm{d}\beta_t + \delta\,\mathrm{d}t.$$

The converse statement is simply the same computation read in reverse on the set where $X_t > 0$. □

The squared form is often the most efficient one for calculations because the singular drift disappears. It makes transparent that the parameter δ is the deterministic rate of growth of the square radius.

The infinitesimal generator of BES^δ on $(0, \infty)$ is

$$\mathcal{L}f(x) = \frac{1}{2}f''(x) + \frac{\delta - 1}{2x}f'(x).$$

This formula packages essentially all of the qualitative behavior of the process.

Proposition 31.3 (Canonical harmonic functions)**.** *For $x > 0$, define*

$$h_\delta(x) = \begin{cases} \log x, & \delta = 2, \\ x^{2-\delta}, & \delta \neq 2. \end{cases}$$

Then $\mathcal{L}h_\delta = 0$ on $(0, \infty)$. Consequently, for every $0 < a < r < b$ and every stopping time

$$\tau_{a,b} := \inf\{t \geq 0 : R_t \in \{a, b\}\},$$

the stopped process $h_\delta(R_{t\wedge\tau_{a,b}})$ is a martingale.

Proof. When $\delta \neq 2$ one computes

$$h'_\delta(x) = (2-\delta)x^{1-\delta}, \qquad h''_\delta(x) = (2-\delta)(1-\delta)x^{-\delta}.$$

Therefore

$$\mathcal{L}h_\delta(x) = \frac{1}{2}(2-\delta)(1-\delta)x^{-\delta} + \frac{\delta-1}{2x}(2-\delta)x^{1-\delta} = 0.$$

For $\delta = 2$, one has

$$h'_2(x) = \frac{1}{x}, \qquad h''_2(x) = -\frac{1}{x^2},$$

and hence

$$\mathcal{L}h_2(x) = \frac{1}{2}\Big(-\frac{1}{x^2}\Big) + \frac{1}{2x}\cdot\frac{1}{x} = 0.$$

Applying Itô's formula to $h_\delta(R_{t\wedge\tau_{a,b}})$ shows that the drift vanishes identically, so the stopped process is a local martingale. Since $R_{t\wedge\tau_{a,b}} \in [a,b]$, the process is bounded; therefore it is a true martingale. □

The preceding harmonic functions play two distinct roles. They are the radial harmonic functions for Brownian motion in dimensions $d = \delta$, and they are also the scale functions for the one-dimensional diffusion R. Both interpretations are mathematically useful: one is geometric, the other probabilistic.

31.3. **Hitting probabilities, scaling, and the origin.**

Proposition 31.4 (Two-sided hitting probabilities)**.** *Let R be a* BES^δ *process started from $r \in (a,b)$, where $0 < a < r < b$. Then*

$$\mathbb{P}_r(\tau_a < \tau_b) = \begin{cases} \dfrac{b^{2-\delta} - r^{2-\delta}}{b^{2-\delta} - a^{2-\delta}}, & \delta \neq 2, \\[2ex] \dfrac{\log b - \log r}{\log b - \log a}, & \delta = 2. \end{cases}$$

Proof. Use the martingale $h_\delta(R_{t\wedge\tau_{a,b}})$ from the previous proposition. Since $R_{\tau_{a,b}} \in \{a,b\}$,

$$h_\delta(r) = \mathbb{E}_r[h_\delta(R_{\tau_{a,b}})] = h_\delta(a)\mathbb{P}_r(\tau_a < \tau_b) + h_\delta(b)\mathbb{P}_r(\tau_b < \tau_a).$$

Solving for $\mathbb{P}_r(\tau_a < \tau_b)$ gives the stated formulas. □

Proposition 31.5 (Brownian scaling for Bessel processes)**.** *Let R be a* BES^δ *process started from $r \geq 0$. For every $c > 0$, the rescaled process*

$$\widetilde{R}_t := c^{-1}R_{c^2t}$$

is again a BES^δ *process, now started from r/c.*

Proof. Define

$$\widetilde{\beta}_t := c^{-1}\beta_{c^2t}.$$

By Brownian scaling, $\widetilde{\beta}$ is a standard Brownian motion. Using the integral form of the Bessel equation,

$$R_{c^2t} = r + \beta_{c^2t} + \frac{\delta-1}{2}\int_0^{c^2t}\frac{\mathrm{d}s}{R_s},$$

and dividing by c, we get

$$\widetilde{R}_t = \frac{r}{c} + \widetilde{\beta}_t + \frac{\delta - 1}{2} \int_0^t \frac{\mathrm{d}u}{\widetilde{R}_u}.$$

Thus $\widetilde{R}$ satisfies the same equation with the same parameter δ. □

The scaling property explains why the qualitative behavior near the origin is controlled by a single critical dimension. A singular drift of order $1/x$ has exactly the same homogeneity as Brownian scaling, so there is a genuine competition between noise and repulsion.

Theorem 31.6 (Boundary classification at the origin)**.** *Let R be a BES^δ process started from $r > 0$, and let*

$$T_0 := \inf\{t \geq 0 : R_t = 0\}.$$

Then the following trichotomy holds.

(i) *If $0 < \delta < 2$, then $\mathbb{P}_r(T_0 < \infty) = 1$.*

(ii) *If $\delta = 2$, then $\mathbb{P}_r(T_0 < \infty) = 0$, but $\inf_{t \geq 0} R_t = 0$ almost surely.*

(iii) *If $\delta > 2$, then $\mathbb{P}_r(T_0 < \infty) = 0$, and in fact*

$$\mathbb{P}_r\Big(\inf_{t \geq 0} R_t \leq a\Big) = \Big(\frac{a}{r}\Big)^{\delta - 2}, \qquad 0 < a \leq r.$$

Proof. Fix $0 < a < r < b$ and let $\tau_{a,b}$ be as above.

If $0 < \delta < 2$, then $2 - \delta > 0$, so letting $a \downarrow 0$ in the hitting formula gives

$$\mathbb{P}_r(T_0 < \tau_b) = \lim_{a \downarrow 0} \mathbb{P}_r(\tau_a < \tau_b) = \frac{b^{2-\delta} - r^{2-\delta}}{b^{2-\delta}} = 1 - \Big(\frac{r}{b}\Big)^{2-\delta}.$$

Now let $b \uparrow \infty$. The right-hand side tends to 1, so $\mathbb{P}_r(T_0 < \infty) = 1$.

If $\delta = 2$, the hitting formula becomes

$$\mathbb{P}_r(\tau_a < \tau_b) = \frac{\log b - \log r}{\log b - \log a}.$$

Letting $b \uparrow \infty$ with $a > 0$ fixed gives $\mathbb{P}_r(\tau_a < \infty) = 1$. Since this holds for every $a > 0$, the process enters every neighborhood of the origin almost surely. On the other hand, the logarithmic martingale prevents actual hitting: if $T_0 < \infty$ with positive probability, then along that event $\log R_{t \wedge T_0} \to -\infty$, which is incompatible with the martingale identity after localization. Hence $T_0 = \infty$ almost surely.

If $\delta > 2$, then $2 - \delta < 0$. Letting $b \uparrow \infty$ in the hitting formula gives

$$\mathbb{P}_r(\tau_a < \infty) = \lim_{b \uparrow \infty} \frac{b^{2-\delta} - r^{2-\delta}}{b^{2-\delta} - a^{2-\delta}} = \frac{r^{2-\delta}}{a^{2-\delta}} = \Big(\frac{a}{r}\Big)^{\delta - 2}.$$

Since this probability tends to 0 as $a \downarrow 0$, the origin is polar and $T_0 = \infty$ almost surely. □

Proposition 31.7 (The cases $\delta = 1$ and $\delta = 3$)**.** *The one-dimensional Bessel process is reflected Brownian motion:*

$$\mathrm{BES}^1 \stackrel{d}{=} |B|.$$

The three-dimensional Bessel process is the radial part of three-dimensional Brownian motion and therefore solves

$$\mathrm{d}R_t = \frac{1}{R_t}\,\mathrm{d}t + \mathrm{d}\beta_t.$$

Proof. For $\delta = 1$, Tanaka's formula gives

$$|B_t| = |B_0| + \int_0^t \operatorname{sgn}(B_s)\, \mathrm{d}B_s + L_t^0(B).$$

Away from the origin the drift term in the Bessel equation vanishes, and the local time term realizes the reflecting boundary behavior at 0. This is exactly the BES^1 dynamics.

The statement for $\delta = 3$ is just the radial Brownian-motion computation already carried out with $d = 3$. □

31.4. **Why Bessel processes belong naturally in these notes.** There is a conceptual reason for including Bessel processes in a manuscript devoted to the original Itô integral. They are built from the Itô formula in the purest possible way. One begins with a multidimensional Brownian motion, composes it with the smooth radial function $x \mapsto |x|$ away from the origin, and discovers that the transformed process satisfies a nontrivial stochastic differential equation whose drift has geometric meaning. The Bessel family is therefore one of the earliest and clearest examples showing that stochastic calculus does not merely integrate with respect to Brownian motion; it creates new diffusions from old ones by transforming observables.

At the same time, Bessel processes reveal several of the structural themes of stochastic calculus in a single model: singular drift, scale functions, localization, optional stopping, boundary classification, and Brownian scaling. For that reason they form a natural bridge between the original Itô papers and later diffusion theory.

32. Feynman–Kac and the probabilistic solution of parabolic equations

The Itô formula can also be read backward. Instead of starting with a stochastic process and differentiating a smooth observable of it, one may start with a partial differential equation and search for a random quantity whose drift vanishes exactly when the equation is satisfied. This is the content of the Feynman–Kac principle. It is one of the most important conceptual descendants of Itô's formula: the differential identity produces martingales, and those martingales in turn encode solutions of parabolic equations as expectations over diffusion paths.

The principle is conceptually simple. If $u = u(t, x)$ solves a suitable backward parabolic equation and X is a diffusion with generator $\mathcal{L}$, then the process

$$\exp\Big(-\int_t^s c(r, X_r)\, \mathrm{d}r\Big) u(s, X_s)$$

should have zero drift precisely when

$$\partial_t u + \mathcal{L}u - cu = 0.$$

If a source term is present, one adds a compensating time integral. Taking expectations then transports terminal data backward in time. From the point of view of the present manuscript, this is a direct continuation of Itô's 1951 program: once one knows how to differentiate $u(s, X_s)$ correctly, analytic representation formulas become martingale identities.

32.1. **Warm-up: the heat equation.** Before inserting a potential or a drift, it is worth isolating the most elementary case. Let $g : \mathbb{R}^d \to \mathbb{R}$ be bounded and continuous, and define

$$P_t g(x) := \mathbb{E}[g(x + B_t)].$$

This is the heat semigroup. The semigroup property follows immediately from the independent increments of Brownian motion, while the Itô formula explains the corresponding differential equation.

Proposition 32.1 (Heat semigroup representation)**.** *Let $T > 0$ and set*

$$u(t, x) := \mathbb{E}[g(x + B_{T-t})], \qquad 0 \le t \le T.$$

If $u \in C^{1,2}([0, T) \times \mathbb{R}^d) \cap C([0, T] \times \mathbb{R}^d)$, then u solves

$$\partial_t u + \frac{1}{2}\Delta u = 0, \qquad u(T, x) = g(x),$$

if and only if

$$u(t, x) = \mathbb{E}[g(x + B_{T-t})].$$

Proof. If u solves the PDE, apply Itô's formula to $u(s, x + B_{s-t})$ for $s \in [t, T]$. One obtains

$$\mathrm{d}u(s, x + B_{s-t}) = \Big(\partial_s u + \frac{1}{2}\Delta u\Big)(s, x + B_{s-t})\ \mathrm{d}s + \nabla u(s, x + B_{s-t}) \cdot \mathrm{d}B_{s-t}.$$

The drift vanishes by the PDE, so the process is a martingale. Taking expectations at time T yields

$$u(t, x) = \mathbb{E}[g(x + B_{T-t})].$$

Conversely, whenever the expectation defines a $C^{1,2}$ function, the same computation shows that its drift must vanish; hence it satisfies the heat equation with the stated terminal condition. □

This first case already contains the main philosophy: a parabolic equation is solved by constructing the right martingale and evaluating it at terminal time.

32.2. **The Brownian Feynman–Kac formula.** We now introduce a potential term. In the Brownian case the relevant operator is

$$\partial_t + \frac{1}{2}\Delta - c(t, x).$$

The exponential factor below is sometimes called the killing weight, because it discounts paths according to how much potential they accumulate.

Theorem 32.2 (Backward Feynman–Kac formula)**.** *Let $T > 0$, let $c : [0, T] \times \mathbb{R}^d \to \mathbb{R}$ be bounded and continuous, and let $g : \mathbb{R}^d \to \mathbb{R}$ be bounded and continuous. Suppose*

$$u \in C^{1,2}([0, T) \times \mathbb{R}^d) \cap C([0, T] \times \mathbb{R}^d)$$

solves

$$\partial_t u(t, x) + \frac{1}{2}\Delta u(t, x) - c(t, x)u(t, x) = 0, \qquad 0 \le t < T,$$

with terminal condition

$$u(T, x) = g(x).$$

Then for every $0 \le t \le T$ *and every* $x \in \mathbb{R}^d$,

$$u(t,x) = \mathbb{E}_x\Big[\exp\Big(-\int_t^T c(s,B_s)\ \mathrm{d}s\Big)g(B_T)\Big],$$

where under $\mathbb{E}_x$ *the Brownian motion starts from* $B_t = x$.

Proof. Fix $t \in [0,T]$ and define for $s \in [t,T]$

$$A_s := \exp\Big(-\int_t^s c(r,B_r)\ \mathrm{d}r\Big), \qquad M_s := A_s u(s,B_s).$$

The process A has finite variation and satisfies

$$\mathrm{d}A_s = -c(s,B_s)A_s\ \mathrm{d}s.$$

By the time-dependent Itô formula,

$$\mathrm{d}u(s,B_s) = \Big(\partial_s u + \frac{1}{2}\Delta u\Big)(s,B_s)\ \mathrm{d}s + \nabla u(s,B_s)\cdot\ \mathrm{d}B_s.$$

Applying the product rule and using that the covariation between a finite-variation process and a semimartingale vanishes, we obtain

$$\mathrm{d}M_s = A_s\ \mathrm{d}u(s,B_s) + u(s,B_s)\ \mathrm{d}A_s = A_s\Big(\partial_s u + \frac{1}{2}\Delta u - cu\Big)(s,B_s)\ \mathrm{d}s + A_s\nabla u(s,B_s)\cdot\ \mathrm{d}B_s.$$

The PDE kills the drift, so

$$M_s = M_t + \int_t^s A_r \nabla u(r,B_r)\cdot\ \mathrm{d}B_r.$$

Because c and u are bounded on $[0,T]\times\mathbb{R}^d$, the factor A_r is bounded above and below by deterministic constants; after localizing the gradient term on compact balls, the stochastic integral is a true martingale. Hence M itself is a martingale. Taking expectations at time T gives

$$u(t,x) = M_t = \mathbb{E}_x[M_T] = \mathbb{E}_x\Big[\exp\Big(-\int_t^T c(r,B_r)\ \mathrm{d}r\Big)u(T,B_T)\Big].$$

Since $u(T,\cdot) = g$, the result follows. □

Corollary 32.3 (Inhomogeneous Brownian formula)**.** *Assume that* u *solves*

$$\partial_t u + \frac{1}{2}\Delta u - cu + f = 0, \qquad u(T,\cdot) = g,$$

where f *is bounded and continuous. Then*

$$\begin{aligned} u(t,x) = {} & \mathbb{E}_x\Big[\exp\Big(-\int_t^T c(r,B_r)\ \mathrm{d}r\Big)g(B_T)\Big] \\ & + \mathbb{E}_x\Big[\int_t^T \exp\Big(-\int_t^s c(r,B_r)\ \mathrm{d}r\Big)f(s,B_s)\ \mathrm{d}s\Big]. \end{aligned}$$

Proof. Define

$$M_s := A_s u(s,B_s) + \int_t^s A_r f(r,B_r)\ \mathrm{d}r.$$

Exactly the same product-rule computation shows that all drift terms cancel. Thus M is a martingale and expectation at time T yields the formula. □

The preceding theorem already implies uniqueness among classical solutions in the indicated class: if two solutions existed with the same terminal condition, their difference would have zero representation and hence would vanish identically.

32.3. **The general diffusion form.** The Brownian case is only the first layer. Let X be the diffusion

$$\mathrm{d}X_s = b(s, X_s)\ \mathrm{d}s + \sigma(s, X_s)\ \mathrm{d}B_s, \qquad X_t = x,$$

with generator

$$\mathcal{L}_s\varphi(x) = \sum_{i=1}^{d} b_i(s,x)\partial_i\varphi(x) + \frac{1}{2}\sum_{i,j=1}^{d} a_{ij}(s,x)\partial_{ij}\varphi(x), \qquad a = \sigma\sigma^\top.$$

Then the same logic yields the corresponding formula for the backward equation

$$\partial_t u + \mathcal{L}_t u - cu + f = 0.$$

Theorem 32.4 (General diffusion Feynman–Kac formula)**.** *Let b and σ be such that the diffusion $X^{t,x}$ exists, and let c, f be bounded continuous functions on $[0,T]\times\mathbb{R}^d$. Suppose*

$$u \in C^{1,2}([0,T)\times\mathbb{R}^d) \cap C([0,T]\times\mathbb{R}^d)$$

solves

$$\partial_t u + \mathcal{L}_t u - cu + f = 0, \qquad u(T,\cdot) = g,$$

with g bounded and continuous. Then

$$\begin{aligned} u(t,x) = \mathbb{E}\Big[\exp\Big(-\int_t^T c(s, X_s^{t,x})\ \mathrm{d}s\Big) g(X_T^{t,x})\Big] \\ + \mathbb{E}\Big[\int_t^T \exp\Big(-\int_t^s c(r, X_r^{t,x})\ \mathrm{d}r\Big) f(s, X_s^{t,x})\ \mathrm{d}s\Big]. \end{aligned}$$

Proof. Write $X_s = X_s^{t,x}$ for brevity and set

$$A_s := \exp\Big(-\int_t^s c(r, X_r)\ \mathrm{d}r\Big), \qquad M_s := A_s u(s, X_s) + \int_t^s A_r f(r, X_r)\ \mathrm{d}r.$$

Itô's formula for $u(s, X_s)$ gives

$$\mathrm{d}u(s, X_s) = \big(\partial_s u + \mathcal{L}_s u\big)(s, X_s)\ \mathrm{d}s + \nabla u(s, X_s)\sigma(s, X_s)\cdot \mathrm{d}B_s.$$

Since $\mathrm{d}A_s = -c(s, X_s)A_s\ \mathrm{d}s$, the product rule yields

$$\begin{aligned} \mathrm{d}\big(A_s u(s, X_s)\big) = A_s\big(\partial_s u + \mathcal{L}_s u - cu\big)(s, X_s)\ \mathrm{d}s \\ + A_s \nabla u(s, X_s)\sigma(s, X_s)\cdot \mathrm{d}B_s. \end{aligned}$$

Adding $A_s f(s, X_s)\ \mathrm{d}s$ and using the PDE, the drift disappears:

$$\mathrm{d}M_s = A_s \nabla u(s, X_s)\sigma(s, X_s)\cdot \mathrm{d}B_s.$$

After the standard localization on compact space-time sets, M is a martingale. Taking expectations at time T gives the result. □

Remark 32.5. *The theorem can be read in two complementary directions. Analytically, it says that a deterministic PDE can be solved by averaging along diffusion paths. Probabilistically, it says that once the correct exponential weight is chosen, the PDE is exactly the condition that turns the weighted observable into a martingale.*

32.4. **A semigroup point of view.** If $f \equiv 0$, the representation can be written formally as

$$u(t, \cdot) = U_{t,T} g,$$

where $U_{t,T}$ is the evolution operator associated with the time-inhomogeneous generator $\mathcal{L}_t - c(t, \cdot)$. In the time-homogeneous case this reduces to a semigroup

$$P_t^c g(x) := \mathbb{E}_x \Big[e^{-\int_0^t c(X_s)\, \mathrm{d}s} g(X_t) \Big],$$

which is the diffusion semigroup modified by the killing potential c. The ordinary heat semigroup corresponds to the special case $c \equiv 0$ and $X = B$.

From this viewpoint the Feynman–Kac formula is not only a representation theorem. It is also a bridge between three languages: stochastic processes, parabolic PDE, and operator theory. The same object can be viewed as a diffusion expectation, as the solution operator of a PDE, or as a semigroup generated by a second-order differential operator with potential.

32.5. **Narrative meaning of the formula.** The analytic meaning of Feynman–Kac is that a deterministic parabolic equation can be solved by averaging over random trajectories. The probabilistic meaning is even more suggestive: the exponential weight

$$\exp\Big(- \int_t^T c(s, X_s)\, \mathrm{d}s \Big)$$

rewards or penalizes paths according to how much potential they accumulate. When $c \equiv 0$, one simply transports the terminal datum by the diffusion and obtains the heat semigroup or, more generally, the Markov semigroup of the diffusion. When $c \neq 0$, the same averaging remains in force, but each path is reweighted by an exponential functional. The resulting expectation interpolates between diffusion, potential theory, and spectral ideas.

In finance the same mechanism reappears after discounting. The Black–Scholes valuation formula is a Feynman–Kac formula for a diffusion with drift, and the discount factor is the exact analogue of the exponential potential weight above. Thus the road from Itô's differential formula to modern derivative pricing is conceptually short: one differentiates correctly, builds a martingale, and reads off a conditional expectation.

The structural point for the present notes is that the original Itô integral was introduced to make sense of expressions such as

$$\int_0^t H_s \, \mathrm{d}B_s.$$

Once that integral exists and Itô's formula has been proved, the stochastic calculus becomes a machine for manufacturing martingales. Feynman–Kac is one of the most elegant manifestations of that machine.

33. EXERCISES

Exercise 33.1 (The Itô isometry and orthogonality revisited)**.** *Let $H, K \in L^2(\Omega \times [0,t])$ be non-anticipating processes such that the Itô integrals*

$$M_t := \int_0^t H_s \, \mathrm{d}B_s, \qquad N_t := \int_0^t K_s \, \mathrm{d}B_s$$

are well defined. Prove that

$$\mathbb{E}[M_t N_t] = \mathbb{E} \int_0^t H_s K_s \, \mathrm{d}s.$$

Deduce in particular that if

$$\mathbb{E} \int_0^t H_s K_s \, \mathrm{d}s = 0,$$

then M_t and N_t are orthogonal in $L^2(\Omega)$.

Solution. We first prove the identity for adapted step processes. Write

$$H_s = \sum_{i=1}^n \xi_{i-1} \, \mathbf{1}_{(t_{i-1},t_i]}(s), \qquad K_s = \sum_{i=1}^n \eta_{i-1} \, \mathbf{1}_{(t_{i-1},t_i]}(s),$$

on a common partition, where ξ_{i-1}, η_{i-1} are $\mathcal{F}_{t_{i-1}}$-measurable and square integrable. Then

$$M_t = \sum_{i=1}^n \xi_{i-1}\big(B_{t_i \wedge t} - B_{t_{i-1} \wedge t}\big), \qquad N_t = \sum_{i=1}^n \eta_{i-1}\big(B_{t_i \wedge t} - B_{t_{i-1} \wedge t}\big).$$

Hence

$$\mathbb{E}[M_t N_t] = \sum_{i,j=1}^n \mathbb{E}\Big[\xi_{i-1}\eta_{j-1}\Delta_i B \, \Delta_j B\Big],$$

where $\Delta_i B := B_{t_i \wedge t} - B_{t_{i-1} \wedge t}$. If $i < j$, then

$$\mathbb{E}\big[\xi_{i-1}\eta_{j-1}\Delta_i B \, \Delta_j B\big] = \mathbb{E}\Big[\xi_{i-1}\Delta_i B \, \mathbb{E}\big(\eta_{j-1}\Delta_j B \mid \mathcal{F}_{t_{j-1}}\big)\Big] = 0,$$

because η_{j-1} is $\mathcal{F}_{t_{j-1}}$-measurable and the increment $\Delta_j B$ has conditional mean zero. The same argument applies when $j < i$. Therefore only the diagonal terms remain:

$$\mathbb{E}[M_t N_t] = \sum_{i=1}^n \mathbb{E}\big[\xi_{i-1}\eta_{i-1}(\Delta_i B)^2\big].$$

Using the conditional second moment of Brownian increments,

$$\mathbb{E}\big[(\Delta_i B)^2 \mid \mathcal{F}_{t_{i-1}}\big] = t_i \wedge t - t_{i-1} \wedge t,$$

we obtain

$$\mathbb{E}[M_t N_t] = \sum_{i=1}^n \mathbb{E}[\xi_{i-1}\eta_{i-1}](t_i \wedge t - t_{i-1} \wedge t) = \mathbb{E} \int_0^t H_s K_s \, \mathrm{d}s.$$

This proves the bilinear isometry on the step class.

Now let $H^{(n)} \to H$ and $K^{(n)} \to K$ in $L^2(\Omega \times [0,t])$, with $H^{(n)}, K^{(n)}$ adapted step processes. By the Itô isometry,

$$\int_0^t H_s^{(n)} \, \mathrm{d}B_s \to M_t, \qquad \int_0^t K_s^{(n)} \, \mathrm{d}B_s \to N_t$$

in $L^2(\Omega)$. Therefore

$$\mathbb{E}[M_t N_t] = \lim_{n\to\infty} \mathbb{E}\Big[\Big(\int_0^t H_s^{(n)}\,\mathrm{d}B_s\Big)\Big(\int_0^t K_s^{(n)}\,\mathrm{d}B_s\Big)\Big] = \lim_{n\to\infty} \mathbb{E}\int_0^t H_s^{(n)} K_s^{(n)}\,\mathrm{d}s.$$

Since $H^{(n)} \to H$ and $K^{(n)} \to K$ in L^2, the products converge in L^1 by Cauchy–Schwarz:

$$\mathbb{E}\int_0^t \Big|H_s^{(n)}K_s^{(n)} - H_s K_s\Big|\,\mathrm{d}s \le \Big\|H^{(n)} - H\Big\|_{H^2}\Big\|K^{(n)}\Big\|_{H^2} + \|H\|_{H^2}\Big\|K^{(n)} - K\Big\|_{H^2} \to 0.$$

Hence

$$\mathbb{E}[M_t N_t] = \mathbb{E}\int_0^t H_s K_s\,\mathrm{d}s.$$

The orthogonality statement is now immediate. □

Exercise 33.2 (The basic quadratic identity)**.** *Prove directly from the Itô formula that*

$$\int_0^t B_s\,\mathrm{d}B_s = \frac{1}{2}(B_t^2 - t).$$

Then compute the expectation and variance of the integral.

Solution. Apply the one-dimensional Itô formula to the function $F(x) = x^2$. Since $F'(x) = 2x$ and $F''(x) = 2$, we get

$$B_t^2 - B_0^2 = \int_0^t 2B_s\,\mathrm{d}B_s + \int_0^t 1\,\mathrm{d}s.$$

Because $B_0 = 0$, this becomes

$$B_t^2 = 2\int_0^t B_s\,\mathrm{d}B_s + t.$$

Rearranging yields

$$\int_0^t B_s\,\mathrm{d}B_s = \frac{1}{2}(B_t^2 - t).$$

This identity already captures the difference between deterministic and stochastic calculus: in ordinary calculus one would expect the differential of x^2 to be $2x\,\mathrm{d}x$, while the Brownian correction contributes the extra term $\mathrm{d}t$.

For the expectation, either take expectations in the identity or use that every square-integrable Itô integral has mean zero. Thus

$$\mathbb{E}\int_0^t B_s\,\mathrm{d}B_s = 0.$$

For the variance, use the isometry:

$$\mathrm{Var}\Big(\int_0^t B_s\,\mathrm{d}B_s\Big) = \mathbb{E}\Big[\Big(\int_0^t B_s\,\mathrm{d}B_s\Big)^2\Big] = \mathbb{E}\int_0^t B_s^2\,\mathrm{d}s = \int_0^t \mathbb{E}[B_s^2]\,\mathrm{d}s.$$

Since $\mathbb{E}[B_s^2] = s$,

$$\mathrm{Var}\Big(\int_0^t B_s\,\mathrm{d}B_s\Big) = \int_0^t s\,\mathrm{d}s = \frac{t^2}{2}.$$

The same value can be recovered from the explicit identity $\frac{1}{2}(B_t^2 - t)$ and the Gaussian fourth moment $\mathbb{E}[B_t^4] = 3t^2$:

$$\mathbb{E}\Big[\frac{1}{4}(B_t^2 - t)^2\Big] = \frac{1}{4}(3t^2 - 2t^2 + t^2) = \frac{t^2}{2}.$$

□

Exercise 33.3 (Exponential martingale)**.** *Fix $\lambda \in \mathbb{R}$ and define*

$$Z_t := \exp\Big(\lambda B_t - \frac{\lambda^2}{2}t\Big), \qquad t \geq 0.$$

Show that $(Z_t)_{t\geq 0}$ is a martingale. Then prove that

$$\mathbb{E}[e^{\lambda B_t}] = e^{\lambda^2 t/2}.$$

Solution. Consider the function

$$f(t,x) = \exp\Big(\lambda x - \frac{\lambda^2}{2}t\Big).$$

Its derivatives are

$$f_t(t,x) = -\frac{\lambda^2}{2}f(t,x), \qquad f_x(t,x) = \lambda f(t,x), \qquad f_{xx}(t,x) = \lambda^2 f(t,x).$$

Apply the time-dependent Itô formula with $X_t = B_t$. Since $\mathrm{d}B_t = 0 \cdot \mathrm{d}t + 1 \cdot \mathrm{d}B_t$, we obtain

$$\mathrm{d}Z_t = \Big(f_t(t,B_t) + \frac{1}{2}f_{xx}(t,B_t)\Big)\,\mathrm{d}t + f_x(t,B_t)\,\mathrm{d}B_t.$$

Substituting the derivatives gives

$$\mathrm{d}Z_t = \Big(-\frac{\lambda^2}{2}f + \frac{1}{2}\lambda^2 f\Big)(t,B_t)\,\mathrm{d}t + \lambda f(t,B_t)\,\mathrm{d}B_t = \lambda Z_t\,\mathrm{d}B_t.$$

So Z_t is an Itô integral added to the constant $Z_0 = 1$:

$$Z_t = 1 + \int_0^t \lambda Z_s\,\mathrm{d}B_s.$$

Provided the integral is square integrable on bounded intervals, this shows that Z is a martingale. One may verify integrability directly from the Gaussian distribution, but there is an even simpler route: for each fixed t,

$$\mathbb{E}[Z_t] = \mathbb{E}\Big[e^{\lambda B_t - \lambda^2 t/2}\Big] = e^{-\lambda^2 t/2}\mathbb{E}[e^{\lambda B_t}].$$

Thus the martingale identity implies $\mathbb{E}[Z_t] = Z_0 = 1$, from which

$$\mathbb{E}[e^{\lambda B_t}] = e^{\lambda^2 t/2}$$

follows immediately.

Conversely, once this Gaussian moment identity is known, it shows in particular that $Z_t \in L^1$ and confirms the martingale property by conditional expectation:

$$\mathbb{E}[Z_t \mid \mathcal{F}_s] = e^{-\lambda^2 t/2}e^{\lambda B_s}\mathbb{E}[e^{\lambda(B_t - B_s)}] = e^{-\lambda^2 t/2}e^{\lambda B_s}e^{\lambda^2(t-s)/2} = Z_s.$$

So Z is indeed a true martingale. □

Exercise 33.4 (Quadratic variation of an Itô integral)**.** *Let*

$$M_t = \int_0^t H_s\,\mathrm{d}B_s,$$

where H is square integrable and non-anticipating. Prove that the quadratic variation of M is

$$\langle M\rangle_t = \int_0^t H_s^2 \, \mathrm{d}s.$$

More precisely, show that for every sequence of partitions π_n of $[0,t]$ whose mesh tends to 0,

$$\sum_{[u,v]\in\pi_n} (M_v - M_u)^2 \longrightarrow \int_0^t H_s^2 \, \mathrm{d}s$$

in probability.

Solution. We first treat the case where H is an adapted step process. Suppose

$$H_s = \sum_{k=1}^m \xi_{k-1} \, \mathbf{1}_{(t_{k-1},t_k]}(s).$$

Then on each interval $[t_{k-1}, t_k]$ we have

$$M_r - M_{t_{k-1}} = \xi_{k-1}(B_r - B_{t_{k-1}}), \qquad t_{k-1} \le r \le t_k.$$

Hence along any refinement of the partition $\{t_0, \dots, t_m\}$,

$$\sum (M_v - M_u)^2 = \sum_{k=1}^m \xi_{k-1}^2 \sum_{[u,v]\subset[t_{k-1},t_k]} (B_v - B_u)^2.$$

By the quadratic variation of Brownian motion,

$$\sum_{[u,v]\subset[t_{k-1},t_k]} (B_v - B_u)^2 \to t_k - t_{k-1}$$

in probability, and therefore

$$\sum (M_v - M_u)^2 \to \sum_{k=1}^m \xi_{k-1}^2 (t_k - t_{k-1}) = \int_0^t H_s^2 \, \mathrm{d}s.$$

So the claim holds for step integrands.

Now take general $H \in H^2$ and choose adapted step processes $H^{(n)}$ such that

$$\mathbb{E} \int_0^t \left| H_s^{(n)} - H_s \right|^2 \, \mathrm{d}s \to 0.$$

Let

$$M_t^{(n)} := \int_0^t H_s^{(n)} \, \mathrm{d}B_s.$$

By the isometry,

$$\mathbb{E}\Big[\sup_{0\le r\le t} \left| M_r^{(n)} - M_r \right|^2\Big] \lesssim \mathbb{E} \int_0^t \left| H_s^{(n)} - H_s \right|^2 \, \mathrm{d}s \to 0$$

by Doob's L^2 inequality applied to the martingale $M^{(n)} - M$. Thus $M^{(n)} \to M$ uniformly on $[0,t]$ in probability.

Fix $\varepsilon > 0$. Write the partition sum for M as

$$\sum (M_v - M_u)^2 = \sum (M_v^{(n)} - M_u^{(n)})^2 + R_n(\pi),$$

where the remainder $R_n(\pi)$ is controlled by

$$|R_n(\pi)| \le 2 \sup_{r \le t} \left| M_r^{(n)} - M_r \right| \sum_{[u,v] \in \pi} \left(\left| M_v^{(n)} - M_u^{(n)} \right| + |M_v - M_u| \right).$$

For fixed n, the first term converges in probability to $\int_0^t (H_s^{(n)})^2 \, \mathrm{d}s$, since the result is already proved for step integrands. Next,

$$\int_0^t (H_s^{(n)})^2 \, \mathrm{d}s \to \int_0^t H_s^2 \, \mathrm{d}s$$

in L^1, because

$$\mathbb{E} \int_0^t \left| (H_s^{(n)})^2 - H_s^2 \right| \, \mathrm{d}s \le \left\| H^{(n)} - H \right\|_{H^2} \left(\left\| H^{(n)} \right\|_{H^2} + \|H\|_{H^2} \right) \to 0.$$

Finally, one chooses n large so that $M^{(n)}$ is uniformly close to M in probability, and then the mesh small so that the partition sum for $M^{(n)}$ is close to its bracket. This proves

$$\sum_{[u,v] \in \pi_n} (M_v - M_u)^2 \to \int_0^t H_s^2 \, \mathrm{d}s$$

in probability.

The quantity on the right is therefore the quadratic variation of M, and by definition it is denoted $\langle M \rangle_t$. □

Exercise 33.5 (The product formula for two Itô processes)**.** *Let*

$$X_t = X_0 + \int_0^t a_s \, \mathrm{d}s + \int_0^t b_s \, \mathrm{d}B_s, \qquad Y_t = Y_0 + \int_0^t c_s \, \mathrm{d}s + \int_0^t d_s \, \mathrm{d}B_s.$$

Prove that

$$\mathrm{d}(X_t Y_t) = X_t \, \mathrm{d}Y_t + Y_t \, \mathrm{d}X_t + b_t d_t \, \mathrm{d}t.$$

Write the integrated form explicitly.

Solution. Apply the two-variable Itô formula to the smooth function

$$F(x, y) = xy.$$

Its partial derivatives are

$$F_x(x, y) = y, \qquad F_y(x, y) = x, \qquad F_{xx}(x, y) = 0, \qquad F_{yy}(x, y) = 0, \qquad F_{xy}(x, y) = 1.$$

The pair (X_t, Y_t) is driven by the same one-dimensional Brownian motion. Hence the general formula gives

$$\mathrm{d}F(X_t, Y_t) = F_x(X_t, Y_t) \, \mathrm{d}X_t + F_y(X_t, Y_t) \, \mathrm{d}Y_t + \frac{1}{2} \left(F_{xx} b_t^2 + 2F_{xy} b_t d_t + F_{yy} d_t^2 \right) \mathrm{d}t.$$

Substituting the derivatives of F yields

$$\mathrm{d}(X_t Y_t) = Y_t \, \mathrm{d}X_t + X_t \, \mathrm{d}Y_t + b_t d_t \, \mathrm{d}t.$$

Since

$$\mathrm{d}X_t = a_t \, \mathrm{d}t + b_t \, \mathrm{d}B_t, \qquad \mathrm{d}Y_t = c_t \, \mathrm{d}t + d_t \, \mathrm{d}B_t,$$

we can also rewrite the right-hand side as

$$\mathrm{d}(X_tY_t) = \big(X_tc_t + Y_ta_t + b_td_t\big)\,\mathrm{d}t + \big(X_td_t + Y_tb_t\big)\,\mathrm{d}B_t.$$

Integrating from 0 to t gives

$$X_tY_t - X_0Y_0 = \int_0^t \big(X_sc_s + Y_sa_s + b_sd_s\big)\,\mathrm{d}s + \int_0^t \big(X_sd_s + Y_sb_s\big)\,\mathrm{d}B_s.$$

This is the stochastic counterpart of the usual product rule, with the extra drift term $b_td_t\ \mathrm{d}t$ produced by quadratic covariation. □

Exercise 33.6 (A hitting-probability computation by martingales)**.** *Fix $a, b > 0$ and define the stopping time*

$$\tau := \inf\{t \geq 0 : B_t = a \text{ or } B_t = -b\}.$$

Show that $\tau < \infty$ almost surely and compute

$$\mathbb{P}\{B_\tau = a\} \qquad \text{and} \qquad \mathbb{P}\{B_\tau = -b\}.$$

Solution. The Brownian motion must hit either a or $-b$ almost surely, so $\tau < \infty$ almost surely. A quick way to see this is to use the recurrence of one-dimensional Brownian motion; however, for the present computation one can also proceed by localization and then verify that the eventual probabilities sum to one.

Since $(B_t)_{t\geq 0}$ is a martingale and the stopped process $B_{t\wedge\tau}$ is bounded between $-b$ and a, the optional stopping theorem applies. Therefore

$$\mathbb{E}[B_\tau] = \mathbb{E}[B_0] = 0.$$

Write

$$p := \mathbb{P}\{B_\tau = a\}.$$

Then

$$\mathbb{P}\{B_\tau = -b\} = 1 - p,$$

and the stopped value satisfies

$$\mathbb{E}[B_\tau] = ap - b(1-p).$$

Since this expectation is zero,

$$ap - b + bp = 0.$$

Hence

$$p(a+b) = b, \qquad \text{extso} \qquad p = \frac{b}{a+b}.$$

Therefore

$$\mathbb{P}\{B_\tau = a\} = \frac{b}{a+b}, \qquad \mathbb{P}\{B_\tau = -b\} = \frac{a}{a+b}.$$

This is the Brownian analogue of the gambler's ruin formula. □

Exercise 33.7 (A heat-equation martingale)**.** *Let $u \in C^{1,2}([0,T]\times\mathbb{R})$ satisfy the backward heat equation*

$$\partial_t u + \frac{1}{2}\partial_{xx}u = 0 \qquad \text{on } [0,T)\times\mathbb{R}.$$

Show that the process

$$M_t := u(t, B_t), \qquad 0 \leq t \leq T,$$

is a martingale. Then deduce that

$$u(t,x) := \mathbb{E}[\varphi(x + B_{T-t})]$$

solves the backward heat equation with terminal condition $u(T,x) = \varphi(x)$, provided φ is bounded and continuous.

Solution. Apply the time-dependent Itô formula to $u(t, B_t)$. Since the driving process is Brownian motion itself, we obtain

$$\mathrm{d}u(t,B_t) = \Big(u_t(t,B_t) + \frac{1}{2}u_{xx}(t,B_t)\Big)\,\mathrm{d}t + u_x(t,B_t)\,\mathrm{d}B_t.$$

Because u satisfies the backward heat equation, the drift term vanishes identically. Thus

$$\mathrm{d}M_t = u_x(t,B_t)\,\mathrm{d}B_t,$$

which shows that M is a local martingale. Under the boundedness assumptions needed to justify integrability, it is in fact a true martingale. In particular,

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s, \qquad 0 \le s \le t \le T.$$

Now define

$$u(t,x) := \mathbb{E}[\varphi(x + B_{T-t})].$$

If we start Brownian motion at x at time t, then the Markov property shows that

$$\nu(t,B_t) = \mathbb{E}[\varphi(B_T) \mid \mathcal{F}_t].$$

The right-hand side is a martingale in t, so by the first part, ν must satisfy the backward heat equation. At terminal time,

$$\nu(T,x) = \mathbb{E}[\varphi(x + B_0)] = \varphi(x).$$

Thus ν solves

$$\partial_t \nu + \frac{1}{2}\partial_{xx}\nu = 0, \qquad \nu(T,x) = \varphi(x).$$

This is one of the first and most important bridges between stochastic calculus and parabolic partial differential equations. □

Exercise 33.8 (Brownian scaling for Itô integrals)**.** *Let $H = (H_t)_{t\ge 0}$ be a square-integrable adapted process and let $c > 0$. Define*

$$\widetilde{B}_t := c^{-1/2}B_{ct}, \qquad \widetilde{H}_t := H_{ct}.$$

Show that $\widetilde{B}$ is a Brownian motion and that for every $t \ge 0$,

$$\int_0^{ct} H_s\,\mathrm{d}B_s = \sqrt{c}\int_0^t \widetilde{H}_u\,\mathrm{d}\widetilde{B}_u.$$

Deduce that if $\tau_a := \inf\{t \ge 0 : B_t = a\}$, then

$$\tau_a \overset{d}{=} a^2\tau_1, \qquad a > 0.$$

Solution. The process $\widetilde{B}$ is centered Gaussian, has continuous paths, and

$$\mathbb{E}[(\widetilde{B}_t - \widetilde{B}_s)^2] = c^{-1}\mathbb{E}[(B_{ct} - B_{cs})^2] = c^{-1}\,c(t-s) = t - s.$$

Moreover the increments are independent because the increments of B over disjoint intervals are independent. Hence $\widetilde{B}$ is again a Brownian motion.

We first verify the scaling identity for step processes. Suppose

$$H_s = \sum_{k=1}^{n} \xi_{k-1} \mathbf{1}_{(t_{k-1},t_k]}(s)$$

on $[0, ct]$, where each ξ_{k-1} is $\mathcal{F}_{t_{k-1}}$-measurable. Then

$$\int_0^{ct} H_s \, \mathrm{d}B_s = \sum_{k=1}^{n} \xi_{k-1}(B_{t_k} - B_{t_{k-1}}).$$

Write $u_k = t_k/c$. Since $B_{t_k} - B_{t_{k-1}} = \sqrt{c}\,(\widetilde{B}_{u_k} - \widetilde{B}_{u_{k-1}})$, we get

$$\int_0^{ct} H_s \, \mathrm{d}B_s = \sqrt{c} \sum_{k=1}^{n} \xi_{k-1}(\widetilde{B}_{u_k} - \widetilde{B}_{u_{k-1}}) = \sqrt{c} \int_0^{t} \widetilde{H}_u \, \mathrm{d}\widetilde{B}_u.$$

The general case follows by approximation in L^2.

Now let $\tau_a := \inf\{t \geq 0 : B_t = a\}$. Since

$$B_t \overset{d}{=} a\widetilde{B}_{t/a^2},$$

we have

$$\{\tau_a \leq t\} = \Big\{\sup_{0\leq s\leq t} B_s \geq a\Big\} = \Big\{\sup_{0\leq u\leq t/a^2} \widetilde{B}_u \geq 1\Big\}.$$

Therefore

$$\mathbb{P}(\tau_a \leq t) = \mathbb{P}(\tau_1 \leq t/a^2),$$

which is exactly the distributional identity $\tau_a \overset{d}{=} a^2\tau_1$. □

Exercise 33.9 (Quadratic variation along dyadic partitions)**.** *Let*

$$Q_n(t) := \sum_{k=0}^{2^n-1} \big(B_{(k+1)t/2^n} - B_{kt/2^n}\big)^2.$$

Show that for every fixed $t \geq 0$,

$$Q_n(t) \to t \qquad \text{in } L^2 \text{ and hence in probability.}$$

Solution. Write

$$\Delta_k^{(n)} B := B_{(k+1)t/2^n} - B_{kt/2^n}.$$

Then the random variables $\Delta_k^{(n)} B$ are independent centered Gaussians with variance $t/2^n$. Hence

$$\mathbb{E}[Q_n(t)] = \sum_{k=0}^{2^n-1} \mathbb{E}[(\Delta_k^{(n)} B)^2] = 2^n \cdot \frac{t}{2^n} = t.$$

To compute the variance, note that if $X \sim N(0, \sigma^2)$ then

$$\mathbb{E}[X^4] = 3\sigma^4, \qquad \mathrm{Var}(X^2) = 3\sigma^4 - \sigma^4 = 2\sigma^4.$$

Therefore

$$\mathrm{Var}(Q_n(t)) = \sum_{k=0}^{2^n-1} \mathrm{Var}\big((\Delta_k^{(n)} B)^2\big) = 2^n \cdot 2\Big(\frac{t}{2^n}\Big)^2 = \frac{2t^2}{2^n}.$$

Thus

$$\mathbb{E}[(Q_n(t) - t)^2] = \mathrm{Var}(Q_n(t)) \to 0.$$

So $Q_n(t) \to t$ in L^2, hence in probability. This is the basic quadratic-variation computation underlying Itô's formula. □

Exercise 33.10 (Reflection principle and the running maximum)**.** *Let $M_t := \sup_{0\le s\le t} B_s$. Show that for every $a > 0$,*

$$\mathbb{P}(M_t \ge a) = 2\mathbb{P}(B_t \ge a).$$

Deduce that

$$M_t \overset{d}{=} |B_t|.$$

Solution. Consider the first hitting time

$$\tau_a := \inf\{s \ge 0 : B_s = a\}.$$

Then

$$\{M_t \ge a\} = \{\tau_a \le t\}.$$

On the event $\{\tau_a \le t\}$, define the reflected path

$$\widetilde{B}_s = \begin{cases} B_s, & 0 \le s \le \tau_a, \\ 2a - B_s, & s \ge \tau_a. \end{cases}$$

By the strong Markov property and symmetry of Brownian increments, $\widetilde{B}$ has the same law as B. Moreover the reflection maps the event $\{B_t \le a,\, M_t \ge a\}$ bijectively onto $\{B_t \ge a\}$. Hence

$$\mathbb{P}(M_t \ge a,\, B_t \le a) = \mathbb{P}(B_t \ge a).$$

Since $\{B_t \ge a\} \subseteq \{M_t \ge a\}$, we get

$$\mathbb{P}(M_t \ge a) = \mathbb{P}(B_t \ge a) + \mathbb{P}(M_t \ge a,\, B_t \le a) = 2\mathbb{P}(B_t \ge a).$$

Now for $x \ge 0$,

$$\mathbb{P}(M_t \le x) = 1 - 2\mathbb{P}(B_t \ge x) = \mathbb{P}(|B_t| \le x),$$

so $M_t \overset{d}{=} |B_t|$. □

Exercise 33.11 (Laplace transform of the first hitting time)**.** *Let $a > 0$ and*

$$\tau_a := \inf\{t \ge 0 : B_t = a\}.$$

Show that for every $\lambda > 0$,

$$\mathbb{E}[e^{-\lambda\tau_a}] = e^{-a\sqrt{2\lambda}}.$$

Deduce that $\tau_a < \infty$ almost surely.

Solution. Fix $\theta > 0$ and consider the exponential martingale

$$N_t := \exp\Big(\theta B_t - \frac{\theta^2}{2}t\Big).$$

For each $n \in \mathbb{N}$, the stopping time $\tau_a \wedge n$ is bounded, so optional stopping gives

$$1 = \mathbb{E}[N_{\tau_a\wedge n}].$$

Since $B_{\tau_a \wedge n} \le a$ on $\{\tau_a > n\}$ and equals a on $\{\tau_a \le n\}$, we have

$$1 = \mathbb{E}\Big[e^{\theta B_{\tau_a \wedge n} - \theta^2(\tau_a \wedge n)/2}\Big].$$

Now choose $\theta = \sqrt{2\lambda}$. On $\{\tau_a \le n\}$ this becomes $e^{a\sqrt{2\lambda}} e^{-\lambda \tau_a}$. On $\{\tau_a > n\}$, the factor is bounded by $e^{\theta B_n - \lambda n}$ and tends to 0 in L^1 after localization; a standard way to avoid this issue is to stop also at τ_{-m} and then let $m \to \infty$. Equivalently, one may apply the same argument to the bounded stopping time $\tau_a \wedge \tau_{-m} \wedge n$ and pass to the limit. The result is

$$1 = e^{a\sqrt{2\lambda}} \mathbb{E}[e^{-\lambda \tau_a}].$$

Hence

$$\mathbb{E}[e^{-\lambda \tau_a}] = e^{-a\sqrt{2\lambda}}.$$

Finally, letting $\lambda \downarrow 0$ yields

$$\mathbb{P}(\tau_a < \infty) = \lim_{\lambda \downarrow 0} \mathbb{E}[e^{-\lambda \tau_a}] = 1.$$

□

Exercise 33.12 (The Brownian bridge)**.** *Define*

$$\beta_t := B_t - tB_1, \qquad 0 \le t \le 1.$$

Show that β is a centered Gaussian process with covariance

$$\mathbb{E}[\beta_s \beta_t] = \min\{s, t\} - st.$$

Show moreover that the process $(\beta_t)_{0 \le t \le 1}$ is independent of B_1.

Solution. Because β_t is a linear combination of Gaussian random variables, the process is Gaussian. It is centered since $\mathbb{E}[B_t] = 0$ and $\mathbb{E}[B_1] = 0$.

Assume $0 \le s \le t \le 1$. Then

$$\begin{aligned}\mathbb{E}[\beta_s \beta_t] &= \mathbb{E}[(B_s - sB_1)(B_t - tB_1)] \\ &= \mathbb{E}[B_s B_t] - t\mathbb{E}[B_s B_1] - s\mathbb{E}[B_t B_1] + st\mathbb{E}[B_1^2].\end{aligned}$$

Since $\mathbb{E}[B_u B_v] = \min\{u, v\}$, this becomes

$$\mathbb{E}[\beta_s \beta_t] = s - ts - st + st = s - st.$$

Since $s = \min\{s, t\}$, we obtain

$$\mathbb{E}[\beta_s \beta_t] = \min\{s, t\} - st.$$

To prove independence from B_1, it is enough in the Gaussian setting to show zero covariance. For each t,

$$\mathbb{E}[\beta_t B_1] = \mathbb{E}[B_t B_1] - t\mathbb{E}[B_1^2] = t - t = 0.$$

Thus every finite vector $(\beta_{t_1}, \dots, \beta_{t_n})$ is jointly Gaussian with B_1 and uncorrelated with it, hence independent of B_1. □

Exercise 33.13 (Conditioning on the endpoint)**.** *Fix $0 \le t \le 1$. Show that*

$$\mathbb{E}[B_t \mid B_1] = tB_1, \qquad \mathrm{Var}(B_t \mid B_1) = t(1-t).$$

Deduce that the conditional law of B_t given $B_1 = y$ is Gaussian with mean ty and variance $t(1-t)$.

Solution. From the previous exercise,

$$B_t = tB_1 + \beta_t,$$

where β_t is independent of B_1, centered, and has variance

$$\mathrm{Var}(\beta_t) = t - t^2 = t(1-t).$$

Therefore

$$\mathbb{E}[B_t \mid B_1] = tB_1 + \mathbb{E}[\beta_t \mid B_1] = tB_1,$$

and

$$\mathrm{Var}(B_t \mid B_1) = \mathrm{Var}(\beta_t \mid B_1) = \mathrm{Var}(\beta_t) = t(1-t).$$

Since $B_t = tB_1 + \beta_t$ and β_t is Gaussian independent of B_1, the conditional law of B_t given $B_1 = y$ is

$$N\big(ty,\, t(1-t)\big).$$

□

Exercise 33.14 (Time inversion)**.** *Define, for $t > 0$,*

$$\widehat{B}_t := tB_{1/t}, \qquad \widehat{B}_0 := 0.$$

Show that $(\widehat{B}_t)_{t\geq 0}$ is a Brownian motion.

Solution. We first compute the covariance. For $0 < s \leq t$,

$$\mathbb{E}[\widehat{B}_s \widehat{B}_t] = st\mathbb{E}[B_{1/s}B_{1/t}] = st\min\{1/s, 1/t\} = st \cdot \frac{1}{t} = s.$$

Thus

$$\mathbb{E}[\widehat{B}_s \widehat{B}_t] = \min\{s, t\}.$$

Since the process is Gaussian as a linear image of B, it has the correct finite-dimensional distributions for Brownian motion.

It remains to check continuity at 0. For $t > 0$,

$$\mathbb{E}[\widehat{B}_t^2] = t^2\mathbb{E}[B_{1/t}^2] = t^2 \cdot \frac{1}{t} = t.$$

Hence $\widehat{B}_t \to 0$ in L^2 as $t \downarrow 0$. Because the finite-dimensional distributions match those of Brownian motion, the Kolmogorov continuity theorem applies exactly as for ordinary Brownian motion, and one obtains a continuous version. Therefore $\widehat{B}$ is a Brownian motion. □

Exercise 33.15 (Tanaka's formula for the positive part)**.** *Show that for every $a \in \mathbb{R}$,*

$$(B_t - a)^+ = (B_0 - a)^+ + \int_0^t \mathbf{1}_{\{B_s > a\}}\, \mathrm{d}B_s + \frac{1}{2}L_t^a,$$

where L_t^a denotes the local time of Brownian motion at level a.

Solution. Apply the one-dimensional Tanaka formula to the semimartingale $X_t := B_t - a$ and the convex function $x \mapsto x^+$. Since $\mathrm{d}X_t = \mathrm{d}B_t$, the formula yields

$$X_t^+ = X_0^+ + \int_0^t \mathbf{1}_{\{X_s>0\}}\,\mathrm{d}X_s + \frac{1}{2}L_t^0(X).$$

Now $X_t^+ = (B_t - a)^+$, $X_0^+ = (B_0 - a)^+$, and $\mathbf{1}_{\{X_s>0\}} = \mathbf{1}_{\{B_s>a\}}$. The local time of X at 0 is exactly the local time of B at a, namely

$$L_t^0(X) = L_t^a(B).$$

Thus

$$(B_t - a)^+ = (B_0 - a)^+ + \int_0^t \mathbf{1}_{\{B_s>a\}}\,\mathrm{d}B_s + \frac{1}{2}L_t^a.$$

□

Exercise 33.16 (Expected local time at the origin)**.** *Show that*

$$\mathbb{E}[L_t^0] = \mathbb{E}[|B_t|] = \sqrt{\frac{2t}{\pi}}.$$

Solution. Apply Tanaka's formula at level 0:

$$|B_t| = \int_0^t \operatorname{sgn}(B_s)\,\mathrm{d}B_s + L_t^0,$$

where $\operatorname{sgn}(x) = \mathbf{1}_{\{x>0\}} - \mathbf{1}_{\{x<0\}}$. The stochastic integral is a martingale with mean zero, so taking expectations gives

$$\mathbb{E}[|B_t|] = \mathbb{E}[L_t^0].$$

Now $B_t \sim N(0,t)$, so $B_t = \sqrt{t}\,Z$ with $Z \sim N(0,1)$. Therefore

$$\mathbb{E}[|B_t|] = \sqrt{t}\,\mathbb{E}[|Z|].$$

A direct Gaussian computation yields

$$\mathbb{E}[|Z|] = \frac{2}{\sqrt{2\pi}}\int_0^\infty x e^{-x^2/2}\,\mathrm{d}x = \sqrt{\frac{2}{\pi}}.$$

Hence

$$\mathbb{E}[L_t^0] = \sqrt{t}\sqrt{\frac{2}{\pi}} = \sqrt{\frac{2t}{\pi}}.$$

□

Exercise 33.17 (A fourth-moment computation)**.** *Use Itô's formula on $x \mapsto x^4$ to prove that*

$$B_t^4 = 4\int_0^t B_s^3\,\mathrm{d}B_s + 6\int_0^t B_s^2\,\mathrm{d}s.$$

Deduce that

$$\mathbb{E}[B_t^4] = 3t^2.$$

Solution. Let $F(x) = x^4$. Then

$$F'(x) = 4x^3, \qquad F''(x) = 12x^2.$$

Itô's formula gives

$$B_t^4 = B_0^4 + \int_0^t 4B_s^3\,\mathrm{d}B_s + \frac{1}{2}\int_0^t 12B_s^2\,\mathrm{d}s.$$

Since $B_0 = 0$, this becomes

$$B_t^4 = 4\int_0^t B_s^3 \, \mathrm{d}B_s + 6\int_0^t B_s^2 \, \mathrm{d}s.$$

Taking expectations, the stochastic integral disappears, and we obtain

$$\mathbb{E}[B_t^4] = 6\int_0^t \mathbb{E}[B_s^2] \, \mathrm{d}s.$$

But $\mathbb{E}[B_s^2] = s$, so

$$\mathbb{E}[B_t^4] = 6\int_0^t s \, \mathrm{d}s = 3t^2.$$

□

Exercise 33.18 (Exit time from a bounded interval)**.** *For $a, b > 0$, let*

$$\tau := \inf\{t \geq 0 : B_t \notin (-a, b)\}.$$

Show that

$$\mathbb{E}[\tau] = ab.$$

More generally, if Brownian motion starts at $x \in (-a, b)$, show that

$$\mathbb{E}_x[\tau] = (b - x)(a + x).$$

Solution. We begin at $x = 0$. The process

$$M_t := B_t^2 - t$$

is a martingale by Itô's formula. Since $\tau \wedge n$ is bounded, optional stopping yields

$$0 = \mathbb{E}[B_{\tau\wedge n}^2 - (\tau \wedge n)].$$

Hence

$$\mathbb{E}[\tau \wedge n] = \mathbb{E}[B_{\tau\wedge n}^2].$$

As $n \to \infty$, we have $B_{\tau\wedge n} \to B_\tau$, and $B_\tau \in \{-a, b\}$. Therefore

$$\mathbb{E}[\tau] = \mathbb{E}[B_\tau^2] = a^2\mathbb{P}(B_\tau = -a) + b^2\mathbb{P}(B_\tau = b).$$

From the gambler's ruin computation,

$$\mathbb{P}(B_\tau = b) = \frac{a}{a+b}, \qquad \mathbb{P}(B_\tau = -a) = \frac{b}{a+b}.$$

Thus

$$\mathbb{E}[\tau] = a^2\frac{b}{a+b} + b^2\frac{a}{a+b} = ab.$$

For a starting point $x \in (-a, b)$, let $u(x) := (b - x)(a + x)$. Then

$$u''(x) = -2, \qquad u(-a) = \nu(b) = 0.$$

Apply Itô's formula up to $t \wedge \tau$:

$$\nu(B_{t\wedge\tau}) = \nu(x) + \int_0^{t\wedge\tau} \nu'(B_s) \, \mathrm{d}B_s + \frac{1}{2}\int_0^{t\wedge\tau} \nu''(B_s) \, \mathrm{d}s.$$

Since $\nu'' = -2$, this becomes

$$\nu(B_{t\wedge\tau}) = \nu(x) + \int_0^{t\wedge\tau} \nu'(B_s)\ \mathrm{d}B_s - (t \wedge \tau).$$

Taking expectations and using $\nu(B_\tau) = 0$, then letting $t \to \infty$, yields

$$\mathbb{E}_x[\tau] = \nu(x) = (b - x)(a + x).$$

□

Exercise 33.19 (Brownian motion with drift and a hitting probability)**.** *Fix $\mu > 0$ and consider*

$$X_t := B_t + \mu t.$$

Let

$$\tau_{-a} := \inf\{t \geq 0 : X_t = -a\}, \qquad a > 0.$$

Show that

$$\mathbb{P}(\tau_{-a} < \infty) = e^{-2\mu a}.$$

Solution. Consider the exponential process

$$N_t := e^{-2\mu X_t} = e^{-2\mu B_t - 2\mu^2 t}.$$

Since

$$-2\mu B_t - 2\mu^2 t = \theta B_t - \frac{\theta^2}{2} t \qquad \text{with } \theta = -2\mu,$$

we see that N_t is an exponential martingale. Let $\tau := \tau_{-a}$. For bounded stopping times $\tau \wedge n$,

$$1 = \mathbb{E}[N_{\tau\wedge n}].$$

On $\{\tau \leq n\}$, we have $X_\tau = -a$, so $N_\tau = e^{2\mu a}$. On $\{\tau > n\}$, one has $X_n > -a$ and therefore $0 < N_n \leq e^{2\mu a} e^{-2\mu(B_n + a) + \cdots}$; the clean argument is to observe that $N_{\tau\wedge n}$ is uniformly integrable here because it is a positive martingale stopped at a bounded time. Passing to the limit gives

$$1 = e^{2\mu a}\mathbb{P}(\tau < \infty) + \mathbb{E}[N_\infty; \tau = \infty].$$

But on $\{\tau = \infty\}$ the drifted Brownian motion tends to $+\infty$ almost surely, so $N_t = e^{-2\mu X_t} \to 0$. Hence

$$1 = e^{2\mu a}\mathbb{P}(\tau < \infty),$$

and therefore

$$\mathbb{P}(\tau_{-a} < \infty) = e^{-2\mu a}.$$

□

Exercise 33.20 (The Ornstein–Uhlenbeck equation)**.** *Let X solve*

$$\mathrm{d}X_t = -\theta X_t\ \mathrm{d}t + \sigma\ \mathrm{d}B_t, \qquad X_0 = x,$$

with $\theta > 0$ and $\sigma > 0$. Solve the equation explicitly and compute $\mathbb{E}[X_t]$ and $\mathrm{Var}(X_t)$.

Solution. Multiply by the integrating factor $e^{\theta t}$. Using Itô's product rule,

$$\mathrm{d}(e^{\theta t} X_t) = \theta e^{\theta t} X_t\ \mathrm{d}t + e^{\theta t}\ \mathrm{d}X_t = e^{\theta t}\sigma\ \mathrm{d}B_t.$$

Integrating from 0 to t gives

$$e^{\theta t} X_t = x + \sigma \int_0^t e^{\theta s} \, \mathrm{d}B_s.$$

Hence

$$X_t = x e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} \, \mathrm{d}B_s.$$

Taking expectations,

$$\mathbb{E}[X_t] = x e^{-\theta t}.$$

For the variance, use the Itô isometry:

$$\mathrm{Var}(X_t) = \sigma^2 \int_0^t e^{-2\theta(t-s)} \, \mathrm{d}s = \sigma^2 \int_0^t e^{-2\theta u} \, \mathrm{d}u = \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t}).$$

Thus

$$X_t \sim N\Big(x e^{-\theta t}, \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t})\Big).$$

□

Exercise 33.21 (A simple Girsanov change of measure)**.** *Fix $T > 0$ and $\theta \in \mathbb{R}$. Define*

$$Z_T := \exp\Big(-\theta B_T - \frac{\theta^2}{2} T\Big), \qquad \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} := Z_T \quad \text{on } \mathcal{F}_T.$$

Show that under $\mathbb{Q}$, the process

$$W_t := B_t + \theta t, \qquad 0 \le t \le T,$$

is a Brownian motion.

Solution. We verify the martingale characterization. First, W has continuous paths because B does. Next, for $0 \le s < t \le T$ and any bounded $\mathcal{F}_s$-measurable random variable H,

$$\mathbb{E}_{\mathbb{Q}}[H(W_t - W_s)] = \mathbb{E}_{\mathbb{P}}\big[H Z_T (B_t - B_s + \theta(t-s))\big].$$

Factor

$$Z_T = Z_s \exp\Big(-\theta(B_t - B_s) - \frac{\theta^2}{2}(t-s)\Big) R_{t,T},$$

where $R_{t,T}$ depends only on the increment after time t and has conditional expectation 1 given $\mathcal{F}_t$. Taking conditional expectations first with respect to $\mathcal{F}_t$, then with respect to $\mathcal{F}_s$, reduces the problem to the centered Gaussian increment $B_t - B_s$. A direct Gaussian computation shows

$$\mathbb{E}\Big[e^{-\theta(B_t - B_s) - \frac{\theta^2}{2}(t-s)}(B_t - B_s + \theta(t-s))\Big] = 0.$$

Hence

$$\mathbb{E}_{\mathbb{Q}}[W_t - W_s \mid \mathcal{F}_s] = 0.$$

A similar computation shows

$$\mathbb{E}_{\mathbb{Q}}[(W_t - W_s)^2 \mid \mathcal{F}_s] = t - s.$$

Because the finite-dimensional distributions are Gaussian, these two identities imply that W is a Brownian motion under $\mathbb{Q}$. □

Exercise 33.22 (A Feynman–Kac exercise with constant potential)**.** *Let $\lambda \geq 0$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be bounded and continuous. Define*

$$u(t,x) := \mathbb{E}\big[e^{-\lambda(T-t)}\varphi(x + B_{T-t})\big], \qquad 0 \leq t \leq T.$$

Show that u solves

$$\partial_t u + \frac{1}{2}\partial_{xx} u - \lambda u = 0, \qquad u(T,x) = \varphi(x).$$

Solution. Write

$$M_s := e^{-\lambda s} u(t+s, x+B_s), \qquad 0 \leq s \leq T-t.$$

Apply Itô's formula:

$$\begin{aligned} \mathrm{d}M_s = e^{-\lambda s}\Big[-\lambda u + (\partial_t u) + \frac{1}{2}\partial_{xx} u\Big](t+s, x+B_s)\ \mathrm{d}s \\ + e^{-\lambda s}(\partial_x u)(t+s, x+B_s)\ \mathrm{d}B_s. \end{aligned}$$

If u satisfies

$$\partial_t u + \frac{1}{2}\partial_{xx} u - \lambda u = 0,$$

then the drift vanishes and M is a martingale. Conversely, starting from the expectation formula and the Markov property, the process M is exactly the conditional expectation of the terminal random variable $e^{-\lambda(T-t)}\varphi(x + B_{T-t})$, hence a martingale. Therefore the drift must vanish, which gives the PDE.

At terminal time,

$$u(T,x) = \mathbb{E}[e^0 \varphi(x + B_0)] = \varphi(x).$$

Thus u solves the stated backward equation. □

Exercise 33.23 (An integral representation for the running maximum)**.** *Let $M_t := \sup_{0 \leq s \leq t} B_s$. Show that for every $a > 0$,*

$$\mathbb{E}[(M_t - a)^+] = \mathbb{E}[(|B_t| - a)^+].$$

Then compute

$$\mathbb{E}[M_t] = \sqrt{\frac{2t}{\pi}}.$$

Solution. From the reflection principle we already know that

$$M_t \overset{d}{=} |B_t|.$$

Therefore every bounded measurable function of M_t has the same expectation as the corresponding function of $|B_t|$, in particular

$$\mathbb{E}[(M_t - a)^+] = \mathbb{E}[(|B_t| - a)^+].$$

Taking $a = 0$ yields

$$\mathbb{E}[M_t] = \mathbb{E}[|B_t|].$$

Since $B_t \sim N(0,t)$,

$$\mathbb{E}[M_t] = \sqrt{t}\,\mathbb{E}[|Z|] = \sqrt{t}\sqrt{\frac{2}{\pi}} = \sqrt{\frac{2t}{\pi}}.$$

□

Exercise 33.24 (A planar Brownian exit-probability exercise)**.** *Let* $(B_t^{(1)}, B_t^{(2)})$ *be a two-dimensional Brownian motion started from* $(x, y) \neq (0, 0)$*, and write*

$$R_t := \sqrt{(B_t^{(1)})^2 + (B_t^{(2)})^2}.$$

Fix $0 < a < \sqrt{x^2 + y^2} < b$*, and let*

$$\tau := \inf\{t \geq 0 : R_t \in \{a, b\}\}.$$

Using the two-dimensional Itô formula applied to $u(z) = \log |z|$*, show that*

$$\mathbb{P}_{(x,y)}\{R_\tau = b\} = \frac{\log \sqrt{x^2 + y^2} - \log a}{\log b - \log a}.$$

Solution. Let $u(z) = \log |z|$ on $\mathbb{R}^2 \setminus \{0\}$. A direct computation shows that $\Delta u = 0$ away from the origin. Therefore, by the two-dimensional Itô formula, the process

$$u(B_{t\wedge\tau}^{(1)}, B_{t\wedge\tau}^{(2)}) = \log R_{t\wedge\tau}$$

is a martingale. Taking expectations and letting $t \to \infty$ gives

$$\log r = \mathbb{E}[\log R_\tau], \qquad r := \sqrt{x^2 + y^2}.$$

Since $R_\tau \in \{a, b\}$, if we write

$$p := \mathbb{P}_{(x,y)}\{R_\tau = b\},$$

then

$$\log r = p \log b + (1 - p) \log a.$$

Solving for p yields

$$p = \frac{\log r - \log a}{\log b - \log a}.$$

This is the logarithmic harmonic measure of the annulus. □

Exercise 33.25 (A planar winding-preparation exercise)**.** *Let* $Z_t = B_t^{(1)} + iB_t^{(2)}$ *be planar Brownian motion started from* $z_0 \neq 0$*, and let*

$$\tau_\varepsilon := \inf\{t \geq 0 : |Z_t| \leq \varepsilon\}$$

for $0 < \varepsilon < |z_0|$*. Show that*

$$\log |Z_{t\wedge\tau_\varepsilon}|$$

is a martingale. Deduce that

$$\mathbb{P}_{z_0}\{\tau_\varepsilon < \tau_R\} = \frac{\log R - \log |z_0|}{\log R - \log \varepsilon}, \qquad R > |z_0|,$$

where $\tau_R := \inf\{t \geq 0 : |Z_t| \geq R\}$*.*

Solution. This is the same harmonic-function argument as in the previous exercise, now written in the stopping-time form used repeatedly in planar Brownian motion. The function $u(z) = \log |z|$ is harmonic on $\{z \neq 0\}$. Hence, by Itô's formula in two dimensions,

$$u(Z_{t\wedge\tau_\varepsilon\wedge\tau_R}) = \log |Z_{t\wedge\tau_\varepsilon\wedge\tau_R}|$$

is a martingale.

Taking expectations and letting $t \to \infty$, we obtain

$$\log |z_0| = \mathbb{E}[\log |Z_{\tau_\varepsilon \wedge \tau_R}|].$$

Since the stopped radius is either ε or R, if

$$p := \mathbb{P}_{z_0}(\tau_\varepsilon < \tau_R),$$

then

$$\log |z_0| = p \log \varepsilon + (1-p) \log R.$$

Solving for p gives

$$p = \frac{\log R - \log |z_0|}{\log R - \log \varepsilon}.$$

Such logarithmic stopping computations are among the elementary ingredients behind deeper winding estimates for planar Brownian motion. □

Exercise 33.26 (Elementary properties of the Cauchy law)**.** *Let X have the standard Cauchy density*

$$f(x) = \frac{1}{\pi(1+x^2)}, \qquad -\infty < x < \infty.$$

(a) *If $a > 0$ and $Y = a^{-1}X$, determine the density of Y.*

(b) *If Y and Z are independent standard Cauchy random variables, show that $(Y+Z)/2$ is again standard Cauchy.*

(c) *For which $r > 0$ is $\mathbb{E}[|X|^r] < \infty$?*

Solution. (a) Since $X = aY$, the change-of-variables formula gives

$$f_Y(y) = a f_X(ay) = \frac{a}{\pi(1+a^2y^2)}, \qquad y \in \mathbb{R}.$$

Thus Y is again Cauchy, with scale parameter $1/a$.

(b) The characteristic function of the standard Cauchy law is

$$\phi_X(\xi) = \mathbb{E}[e^{i\xi X}] = e^{-|\xi|}.$$

Therefore, by independence,

$$\phi_{(Y+Z)/2}(\xi) = \phi_Y(\xi/2)\phi_Z(\xi/2) = e^{-|\xi|/2}e^{-|\xi|/2} = e^{-|\xi|}.$$

This is exactly the characteristic function of the standard Cauchy law.

(c) Near the origin there is no integrability problem, because f is bounded. At infinity,

$$\frac{|x|^r}{\pi(1+x^2)} \sim \frac{1}{\pi}\,|x|^{r-2}.$$

Hence the integral

$$\int_1^\infty x^{r-2}\,\mathrm{d}x$$

converges if and only if $r < 1$. Therefore

$$\mathbb{E}[|X|^r] < \infty \quad \Longleftrightarrow \quad 0 < r < 1.$$

For $r \geq 1$ the absolute moment diverges.

□

Exercise 33.27 (A Cauchy process obtained from planar Brownian motion)**.** *Let* $X_t = (X_t^1, X_t^2)$ *be a standard two-dimensional Brownian motion, and define*

$$\sigma_t := \inf\{s \geq 0 : X_s^2 = t\}, \qquad Y_t := X_{\sigma_t}^1.$$

(a) *Which of the following properties hold?*

(i) $Y_0 = 0$*;*

(ii) *for* $s_1 \leq t_1 \leq s_2 \leq t_2 \leq \cdots \leq s_n \leq t_n$*, the random variables*

$$Y_{t_1} - Y_{s_1}, \ldots, Y_{t_n} - Y_{s_n}$$

are independent;

(iii) *if* $0 \leq s < t$*, then the law of* $Y_t - Y_s$ *is the same as that of* Y_{t-s}*;*

(iv) Y_t *is a continuous function of* t*.*

(b) *For which* $\lambda > 0$ *does the rescaled process*

$$Z_t := a^{-\lambda} Y_{at}$$

have the same finite-dimensional distributions as Y_t*?*

Solution. Let $\tau_t := \sigma_t$ for brevity. Since X^1 and X^2 are independent one-dimensional Brownian motions, the process τ_t is the first-passage process of a Brownian motion and Y is obtained by subordinating an independent Brownian motion by τ.

For $u \geq 0$, the Laplace transform of τ_t is the classical first-passage formula

$$\mathbb{E}[e^{-u\tau_t}] = e^{-t\sqrt{2u}}.$$

Hence, conditioning on τ_t,

$$\mathbb{E}[e^{i\xi Y_t}] = \mathbb{E}\big[e^{-\xi^2 \tau_t/2}\big] = e^{-t|\xi|}.$$

So Y_t has the centered Cauchy distribution with scale parameter t.

(a) (i) Since $\tau_0 = 0$ almost surely and $X_0^1 = 0$, one has $Y_0 = 0$.

(ii) The process $t \mapsto \tau_t$ has independent increments by the strong Markov property applied to X^2 at the stopping times $\tau_{s_1}, \tau_{s_2}, \ldots$. Since X^1 has independent increments and is independent of X^2, the increments

$$Y_{t_k} - Y_{s_k} = X_{\tau_{t_k}}^1 - X_{\tau_{s_k}}^1$$

are independent as well.

(iii) By the same strong Markov argument, $\tau_t - \tau_s$ has the same law as τ_{t-s}, and it is independent of the past up to time τ_s. Therefore

$$Y_t - Y_s = X_{\tau_t}^1 - X_{\tau_s}^1$$

has the same law as $X_{\tau_{t-s}}^1 = Y_{t-s}$.

(iv) This property fails. Indeed, the characteristic function

$$\mathbb{E}[e^{i\xi Y_t}] = e^{-t|\xi|}$$

shows that Y is the standard Cauchy Lévy process. A process with stationary independent increments and continuous paths must be Gaussian, whereas the one-time marginals of Y are Cauchy. Thus Y cannot have continuous sample paths.

Therefore (i), (ii), and (iii) hold, while (iv) does not.

(b) For fixed t and $\xi \in \mathbb{R}$,

$$\mathbb{E}[e^{i\xi Z_t}] = \mathbb{E}[e^{i\xi a^{-\lambda} Y_{at}}] = e^{-at|a^{-\lambda}\xi|} = e^{-t|\xi|a^{1-\lambda}}.$$

For this to equal the characteristic function $e^{-t|\xi|}$ of Y_t for every t and every ξ, we must have

$$a^{1-\lambda} = 1.$$

Hence the only possible value is

$$\lambda = 1.$$

This is exactly the self-similarity exponent of the Cauchy process.

□

Exercise 33.28 (A harmonic observable of multidimensional Brownian motion)**.** *Let $d > 1$ and let W_t be a standard d-dimensional Brownian motion started from $x \neq 0$. Define*

$$M_t = \begin{cases} \log|W_t|, & d = 2, \\ |W_t|^{2-d}, & d > 2. \end{cases}$$

Show that M_t is a local martingale. In dimension 2, show that it is a true martingale.

Solution. Set $u(x) = \log|x|$ when $d = 2$ and $u(x) = |x|^{2-d}$ when $d > 2$. For a radial function $u(x) = g(|x|)$, one has

$$\Delta u(x) = g''(r) + \frac{d-1}{r} g'(r), \qquad r = |x| > 0.$$

In dimension 2, $g(r) = \log r$, so

$$g'(r) = \frac{1}{r}, \qquad g''(r) = -\frac{1}{r^2}, \qquad g''(r) + \frac{1}{r} g'(r) = 0.$$

In dimension $d > 2$, $g(r) = r^{2-d}$, so

$$g'(r) = (2-d)r^{1-d}, \qquad g''(r) = (2-d)(1-d)r^{-d},$$

and therefore

$$g''(r) + \frac{d-1}{r} g'(r) = 0.$$

Thus in both cases u is harmonic on $\mathbb{R}^d \setminus \{0\}$.

Let

$$\tau_n := \inf\{t \geq 0 : |W_t| \notin (1/n, n)\}.$$

Applying the multidimensional Itô formula to $u(W_{t\wedge\tau_n})$ yields

$$u(W_{t\wedge\tau_n}) = u(x) + \sum_{i=1}^{d} \int_0^{t\wedge\tau_n} \partial_i u(W_s)\, \mathrm{d}W_s^i + \frac{1}{2} \int_0^{t\wedge\tau_n} \Delta u(W_s)\, \mathrm{d}s.$$

Because $\Delta u = 0$ on the annulus $\{1/n < |x| < n\}$, the drift term vanishes, so

$$u(W_{t\wedge\tau_n}) = u(x) + \sum_{i=1}^{d} \int_0^{t\wedge\tau_n} \partial_i u(W_s)\, \mathrm{d}W_s^i$$

is a martingale. Letting $n \to \infty$ gives that $M_t = u(W_t)$ is a local martingale.

In dimension 2, a single point is polar for planar Brownian motion, so $W_t \neq 0$ almost surely for every fixed $t > 0$. Moreover,

$$\mathbb{E}\big[|\log |W_t||\big] < \infty,$$

which allows one to pass to the limit in L^1 along the localization. Hence

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s, \qquad 0 \le s \le t,$$

so $M_t = \log |W_t|$ is a true martingale in dimension 2.

For $d > 2$, the same computation always gives a local martingale. One should not automatically expect a true martingale here, because the positive local martingale $|W_t|^{2-d}$ may fail to preserve expectation globally. □

Exercise 33.29 (Exit from an interval for Brownian motion)**.** *Let W_t be a standard one-dimensional Brownian motion and let $a, b > 0$. Define*

$$T_{a,-b} := \inf\{t \ge 0 : W_t = a \text{ or } W_t = -b\}.$$

(a) *Use the martingale W_t to compute $\mathbb{P}(W_{T_{a,-b}} = a)$.*

(b) *Use the martingale $W_t^2 - t$ to compute $\mathbb{E}[T_{a,-b}]$.*

(c) *Show that $T_{a,-a}$ and $W_{T_{a,-a}}$ are independent.*

(d) *Decide whether $T_{a,-b}$ and $W_{T_{a,-b}}$ are independent for arbitrary $a, b > 0$.*

(e) *Compute the Laplace transform of $T_{a,-a}$ and deduce its moment generating function on its interval of finiteness.*

Solution. Write $T := T_{a,-b}$.

(a) Since $W_{t\wedge T}$ is a bounded martingale,

$$0 = \mathbb{E}[W_T] = a\mathbb{P}(W_T = a) - b\mathbb{P}(W_T = -b).$$

Because W_T can take only the two values a and $-b$, we get

$$\mathbb{P}(W_T = a) = \frac{b}{a+b}, \qquad \mathbb{P}(W_T = -b) = \frac{a}{a+b}.$$

(b) The stopped process $W_{t\wedge T}^2 - (t \wedge T)$ is again a martingale. Taking expectations and letting $t \to \infty$ gives

$$\mathbb{E}[T] = \mathbb{E}[W_T^2] = a^2\mathbb{P}(W_T = a) + b^2\mathbb{P}(W_T = -b).$$

Substituting the probabilities from part (a),

$$\mathbb{E}[T] = a^2 \frac{b}{a+b} + b^2 \frac{a}{a+b} = ab.$$

(c) When the interval is symmetric, $T := T_{a,-a}$ is invariant under path reflection $W \mapsto -W$, whereas W_T changes sign. Therefore, for every Borel set $A \subset [0,\infty)$,

$$\mathbb{P}(T \in A, W_T = a) = \mathbb{P}(T \in A, W_T = -a).$$

Since the two events partition $\{T \in A\}$,

$$\mathbb{P}(T \in A, W_T = a) = \frac{1}{2}\mathbb{P}(T \in A) = \mathbb{P}(T \in A)\mathbb{P}(W_T = a).$$

The same identity holds with $-a$ in place of a, so $T_{a,-a}$ and $W_{T_{a,-a}}$ are independent.

(d) In general the answer is no. Let

$$A_\theta := \mathbb{E}\big[e^{-\theta T}; W_T = a\big], \qquad B_\theta := \mathbb{E}\big[e^{-\theta T}; W_T = -b\big], \qquad \theta > 0.$$

Applying optional stopping to the exponential martingales

$$\exp\big(\pm\sqrt{2\theta}\, W_t - \theta t\big)$$

at time T yields the linear system

$$\begin{cases} 1 = e^{\sqrt{2\theta}a} A_\theta + e^{-\sqrt{2\theta}b} B_\theta, \\ 1 = e^{-\sqrt{2\theta}a} A_\theta + e^{\sqrt{2\theta}b} B_\theta. \end{cases}$$

Solving gives

$$A_\theta = \frac{\sinh(\sqrt{2\theta}\, b)}{\sinh(\sqrt{2\theta}(a+b))}, \qquad B_\theta = \frac{\sinh(\sqrt{2\theta}\, a)}{\sinh(\sqrt{2\theta}(a+b))}.$$

Hence

$$\frac{A_\theta}{B_\theta} = \frac{\sinh(\sqrt{2\theta}\, b)}{\sinh(\sqrt{2\theta}\, a)},$$

which depends on θ unless $a = b$. If T and W_T were independent, this ratio would have to be the constant

$$\frac{\mathbb{P}(W_T = a)}{\mathbb{P}(W_T = -b)} = \frac{b}{a}.$$

Therefore independence fails unless the interval is symmetric.

(e) Now take $b = a$ and write $T := T_{a,-a}$. By part (c), T and W_T are independent, and $\mathbb{P}(W_T = a) = \mathbb{P}(W_T = -a) = 1/2$. Optional stopping for the martingale

$$\exp\Big(\lambda W_t - \frac{\lambda^2}{2} t\Big)$$

gives

$$1 = \mathbb{E}\Big[e^{\lambda W_T - \lambda^2 T/2}\Big] = \mathbb{E}\big[e^{-\lambda^2 T/2}\big]\, \mathbb{E}[e^{\lambda W_T}] = \mathbb{E}\big[e^{-\lambda^2 T/2}\big] \cosh(\lambda a).$$

Hence, with $\theta = \lambda^2/2$,

$$\mathbb{E}[e^{-\theta T_{a,-a}}] = \frac{1}{\cosh(a\sqrt{2\theta})}, \qquad \theta \geq 0.$$

This is the Laplace transform. Replacing θ by $-u$ gives the moment generating function on the range where it is finite:

$$\mathbb{E}[e^{uT_{a,-a}}] = \frac{1}{\cos(a\sqrt{2u})}, \qquad 0 \le u < \frac{\pi^2}{8a^2}.$$

□

Exercise 33.30 (The density of a lognormal random variable)**.** *Let N be a standard normal random variable and let*

$$X = ae^{bN}, \qquad a, b > 0.$$

Show that X has density

$$f_X(x) = \frac{1}{bx\sqrt{2\pi}} \exp\Big(-\frac{(\log(x/a))^2}{2b^2}\Big), \qquad x > 0.$$

Solution. The map $n \mapsto x = ae^{bn}$ is a smooth increasing bijection from $\mathbb{R}$ onto $(0, \infty)$, with inverse

$$n = \frac{\log(x/a)}{b}.$$

Its derivative is

$$\frac{\mathrm{d}n}{\mathrm{d}x} = \frac{1}{bx}.$$

Since the density of N is $(2\pi)^{-1/2}e^{-n^2/2}$, the change-of-variables formula gives

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\Big(-\frac{1}{2}\Big(\frac{\log(x/a)}{b}\Big)^2\Big)\frac{1}{bx}, \qquad x > 0.$$

This is exactly the claimed density. □

Exercise 33.31 (A Bessel-type diffusion)**.** *Let W_t be a standard one-dimensional Brownian motion. Suppose $Z_0 = 1$ and that, for $t < T := \inf\{s \ge 0 : Z_s = 0\}$, the process Z_t satisfies*

$$\mathrm{d}Z_t = \frac{a}{Z_t}\,\mathrm{d}t + \mathrm{d}W_t.$$

(a) *Find a nonconstant differentiable function ϕ such that $M_t := \phi(Z_{t\wedge T})$ is a martingale.*

(b) *If $0 < \varepsilon < 1 < \alpha$ and $S = S(\varepsilon, \alpha)$ is the first time t such that $Z_t = \varepsilon$ or $Z_t = \alpha$, compute $\mathbb{P}(Z_S = \varepsilon)$.*

(c) *Compute the probability that there exists some time t such that $Z_t = \varepsilon$. For which values of a is this probability equal to one?*

(d) *For which values of a does the process reach the origin in finite time?*

Solution. This is the Bessel equation written with the parameter $a = (\delta - 1)/2$. The scale function is obtained directly from Itô's formula.

(a) Let $\phi \in C^2((0, \infty))$. Applying Itô's formula to $\phi(Z_t)$ gives

$$\mathrm{d}\phi(Z_t) = \Big(\frac{1}{2}\phi''(Z_t) + \frac{a}{Z_t}\phi'(Z_t)\Big)\,\mathrm{d}t + \phi'(Z_t)\,\mathrm{d}W_t.$$

To kill the drift we require

$$\frac{1}{2}\phi''(x) + \frac{a}{x}\phi'(x) = 0.$$

Equivalently,

$$\phi''(x) + \frac{2a}{x}\phi'(x) = 0.$$

Multiplying by x^{2a} yields

$$\big(x^{2a}\phi'(x)\big)' = 0,$$

so

$$\phi'(x) = Cx^{-2a}.$$

Therefore one may take

$$\phi(x) = \begin{cases} x^{1-2a}, & a \neq \frac{1}{2}, \\ \log x, & a = \frac{1}{2}. \end{cases}$$

Any nonconstant affine multiple of these functions also works.

(b) Let $p = \mathbb{P}(Z_S = \varepsilon)$. Since $\phi(Z_{t\wedge S})$ is bounded, optional stopping gives

$$\phi(1) = \mathbb{E}[\phi(Z_S)] = p\phi(\varepsilon) + (1-p)\phi(\alpha).$$

Thus

$$p = \frac{\phi(\alpha) - \phi(1)}{\phi(\alpha) - \phi(\varepsilon)}.$$

Hence

$$\mathbb{P}(Z_S = \varepsilon) = \begin{cases} \dfrac{\alpha^{1-2a} - 1}{\alpha^{1-2a} - \varepsilon^{1-2a}}, & a \neq \frac{1}{2}, \\ \dfrac{\log \alpha}{\log(\alpha/\varepsilon)}, & a = \frac{1}{2}. \end{cases}$$

(c) Let $\alpha \to \infty$. The event $\{Z_S = \varepsilon\}$ then increases to the event that the process ever hits ε. Therefore

$$\mathbb{P}(\exists t \geq 0 : Z_t = \varepsilon) = \begin{cases} 1, & a < \frac{1}{2}, \\ 1, & a = \frac{1}{2}, \\ \varepsilon^{2a-1}, & a > \frac{1}{2}. \end{cases}$$

So the probability is equal to one exactly when $a \leq \frac{1}{2}$.

(d) In the Bessel parameter $\delta = 2a + 1$, the diffusion is BES^δ. The classical boundary classification says that the origin is reached in finite time if and only if $\delta < 2$, that is,

$$a < \frac{1}{2}.$$

At the critical value $a = 1/2$ the process comes arbitrarily close to the origin almost surely but does not actually hit it, and for $a > 1/2$ the origin is polar.

□

Exercise 33.32 (A Black–Scholes change of unknown)**.** *Suppose that $V = V(t,x)$ satisfies*

$$V_t + \frac{1}{2}\sigma^2 x^2 V_{xx} = 0, \qquad 0 \leq t < T,$$

and define

$$\widetilde{V}(t,x) := e^{r(t-T)}V\big(t, e^{r(T-t)}x\big).$$

Show that $\widetilde{V}$ satisfies the Black–Scholes equation

$$\widetilde{V}_t + rx\widetilde{V}_x + \frac{1}{2}\sigma^2 x^2 \widetilde{V}_{xx} - r\widetilde{V} = 0.$$

Solution. Set

$$y = e^{r(T-t)}x.$$

Then

$$\widetilde{V}(t,x) = e^{r(t-T)}V(t,y).$$

Differentiate carefully. Since $y_t = -ry$ and $y_x = e^{r(T-t)}$, we obtain

$$\widetilde{V}_t = r\widetilde{V} + e^{r(t-T)}\big(V_t - ryV_x\big),$$

$$\widetilde{V}_x = e^{r(t-T)}V_x\, y_x = V_x(t,y),$$

and therefore

$$\widetilde{V}_{xx} = V_{xx}(t,y)e^{r(T-t)}.$$

Multiplying by x^2 gives

$$x^2\widetilde{V}_{xx} = x^2 e^{r(T-t)}V_{xx}(t,y) = e^{-r(T-t)}y^2 V_{xx}(t,y).$$

Hence

$$\frac{1}{2}\sigma^2 x^2 \widetilde{V}_{xx} = e^{r(t-T)}\frac{1}{2}\sigma^2 y^2 V_{xx}(t,y).$$

Also,

$$rx\widetilde{V}_x = rxe^{r(T-t)}V_x(t,y) = ryV_x(t,y)e^{r(t-T)}.$$

Adding the three pieces,

$$\widetilde{V}_t + rx\widetilde{V}_x + \frac{1}{2}\sigma^2 x^2 \widetilde{V}_{xx} - r\widetilde{V} = e^{r(t-T)}\Big(V_t + \frac{1}{2}\sigma^2 y^2 V_{yy}\Big)(t,y).$$

The right-hand side vanishes because V solves the stated equation. Therefore $\widetilde{V}$ satisfies the Black–Scholes PDE. □

34. Concluding perspective

The originality of Itô's theory can be summarized in a single sentence: the stochastic integral is not built by pathwise variation estimates but by Hilbert-space orthogonality. Once that viewpoint is adopted, everything else falls into place. The integral is first defined on step processes, where each increment is visibly meaningful. The L^2 isometry shows that the definition is stable under refinement and makes possible a completion argument. The maximal inequality then upgrades fixed-time L^2 convergence to the existence of continuous versions. Finally, the nontrivial quadratic variation of Brownian motion forces an additional second-order term in the chain rule.

It is this additional term that makes the stochastic differential formula both subtle and powerful. In deterministic calculus the chain rule is first order; in stochastic calculus the chain rule remembers second-order information because the noise itself carries a second-order

mass. That is the core mathematical discovery of the 1944 and 1951 papers, and it remains the conceptual heart of stochastic analysis.

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